Robust error-minimization in the genetic code across physicochemical metrics and variant codes: a graph-theoretic analysis in GF(2)^6

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Abstract

The standard genetic code is structured to reduce the impact of point mutations, but the robustness of this property across physicochemical metrics, naturally variant codes, and codon-reassignment mechanisms remains incompletely quantified. We embed the 64 codons in $\mathrm{GF}(2)^6$, representing the hypercube Q_6 as a coordinate-dependent subgraph of the encoding-independent single-nucleotide mutation graph $H(3,4)$. This framework allows interpolation between Q_6 and the full mutation graph and permits simultaneous analysis of physicochemical error minimization and codon-family topology. Under a block-preserving null model (n =10,000), the standard code is significantly low-cost across four amino-acid distance metrics: Grantham, Miyata, Woese polar requirement, and Kyte–Doolittle hydropathy ($p \leq 0.006$ for all metrics). The error-minimization signal strengthens monotonically as diagonal-edge weight increases from Q_6 toward $H(3,4)$. Across NCBI variant translation tables, this near-optimality is broadly preserved: among the 12 codes differing from the standard code by at least three reassignments, 11 retain top-5% placement after BH–FDR correction, with yeast mitochondrial code as the sole marginal exception; near-standard codes are also significant but are interpreted primarily as confirmations of standard-code geometry. Natural codon reassignments are strongly depleted for topology-breaking moves. Under $H(3,4)$, only 6 of 28 observed events increase amino-acid codon-family disconnectedness, compared with 66% of 1,280 candidate single-codon relabelings (RR 0.32, permutation $p \leq 10^{-4}$). This depletion is robust to alternative topology definitions, phylogenetic clade exclusions, and base-to-bit encodings, motivating $H(3,4)$ as the primary topology graph. Conditional-logit models show that topology avoidance and local physicochemical cost provide complementary explanatory signal, whereas a heuristic tRNA-distance proxy does not improve fit. Several variant-code lineages show suggestive enrichment of tRNA genes for reassigned amino acids, although topology-breaking subsets remain underpowered. Retrospective analysis of published genome-recoding datasets suggests that codon-family topology constrains natural code evolution, whereas acute synthetic-recoding burden is better captured by local neighborhood mismatch and mutational accessibility. Together, these results support a two-axis model of genetic-code evolution in which physicochemical smoothness and codon-family topological integrity impose partly independent constraints.

Keywords: genetic code evolution; error minimization; GF(2)^6 representation; codon reassignment; physicochemical distance; codon-family topology; conditional logit; tRNA gene enrichment; genome recoding

1 Introduction

The standard genetic code maps 61 sense codons to 20 amino acids through a pattern that has long been recognized as non-random. Woese (1965) first noted that similar codons tend to encode amino acids with similar physicochemical properties, and Freeland & Hurst (1998) demonstrated quantitatively that the standard code sits in approximately the top 10^{-6} percentile of random codes for mutational error minimization under the Woese polar requirement distance. These findings established that the code's structure reduces the fitness impact of point mutations, but the question of whether this optimality is specific to particular physicochemical metrics or represents a robust property across several distinct physicochemical parameterizations has not been systematically addressed.

Every codon comprises three nucleotides drawn from $\{C, U, A, G\}$. By choosing a bijection $\varphi : \{C, U, A, G\} \to \mathrm{GF}(2)^2$—for instance, $C \mapsto (0, 0)$, $U \mapsto (0, 1)$, $A \mapsto (1, 0)$, $G \mapsto (1, 1)$—each codon maps to a vertex of the 6-dimensional binary hypercube $Q_6 = \mathrm{GF}(2)^6$. The genetic code then becomes a *coloring* of Q_6 by 21 labels (20 amino acids plus the stop signal), and single-nucleotide mutations correspond to edges of Q_6 or, more precisely, to a subgraph of the complete mutation graph $H(3, 4)$.

This representation is not new in principle—binary encodings of the genetic code appear in the mathematical biology literature (e.g., (Antoneli & Forger, 2011)). Its value lies not in the encoding itself but in the analytical decomposition it enables: the complete single-nucleotide mutation graph $H(3, 4)$ (288 edges) separates into 192 Hamming-distance-1 edges (single-bit changes) and 96 within-nucleotide distance-2 edges (both bits of one nucleotide position flip simultaneously), allowing systematic interpolation via a weight parameter $\rho \in [0, 1]$. Note that this decomposition does not correspond to the biological transition/transversion partition: under 16 of the 24 possible 2-bit encodings, the Hamming-1 edges contain an equal mixture of transitions and transversions per nucleotide position (2 of each among 4 Hamming-1 pairs), while the remaining 8 encodings place both transitions on diagonal (Hamming-2) edges. The ρ parameter should therefore be interpreted as a diagonal-edge inclusion weight, not a transition/transversion weight. Previous work has not exploited this decomposition to test error-minimization across multiple physicochemical metrics, nor extended the analysis to variant genetic codes. We address three questions:

1. **Is the standard code optimal under multiple physicochemical metrics?** We test whether the standard code's edge-mismatch score is extreme relative to block-preserving null models across four distinct physicochemical distance metrics (Grantham, Miyata, Woese polar requirement, and Kyte–Doolittle hydropathy), extending Freeland & Hurst (1998) from a single metric to a cross-metric robustness envelope.

2. **Is this structure preserved by evolution?** We ask whether variant genetic codes (NCBI translation tables 2–33) maintain error-minimization, and whether natural codon reassignment events preferentially avoid disrupting the topological connectivity of amino acid codon families.

3. **What are the genomic correlates of disruption?** We test whether organisms whose variant codes break the connectivity of an amino acid's codon graph show elevated tRNA gene copy numbers for the affected amino acid, using tRNAscan-SE–verified data from 18 genomes spanning 5 variant genetic codes across Alveolata, Opisthokonta, Excavata, and Mollicutes.

To delimit the framework's algebraic scope, we examine four conjectures that have been previously circulated in informal methodological notes and presentations associated with this project (one author's prior unpublished technical note (Clayworth, 2026) the relevant claim statements are quoted verbatim where they are tested)—a Serine distance-4 invariance claim, PSL(2,7) symmetry, a holomorphic embedding extending a character of $\mathrm{GF}(8)^*$, and a KRAS–Fano clinical prediction—and report their falsification or rejection transparently (Section 3.10 and Supplementary Note S1). Including these tests is intended to delimit scope, not to establish straw targets: each conjecture had a coherent algebraic motivation, and showing where the framework does **not** extend is part of giving an honest account of where it does.

The paper is organized as follows. Section 2 describes the encoding formalism, graph decomposition, null models, and statistical methods, including a new retrospective cross-study reanalysis of nine published genome recoding datasets (with quantitative analysis of eight). Section 3 presents the four supported findings (cross-metric coloring optimality, per-table preservation, ρ-robustness, and topology avoidance), the suggestive tRNA enrichment result, the cross-study reanalysis of synthetic recoding outcomes, and exploratory observations. Section 4 discusses the graph-theoretic interpretation, the relationship to frozen-accident versus adaptive hypotheses (Koonin & Novozhilov, 2009), and an exploratory three-layer interpretation of how codon-space structure relates to recoding outcomes. All analyses are reproducible via the open-source `codon-topo` pipeline (version 0.4.0, seed 135325).

2 Methods

2.1 Binary encoding of the genetic code

We encode each nucleotide base as a 2-bit vector via the default bijection $\varphi : C \mapsto (0,0)$, $U \mapsto (0,1)$, $A \mapsto (1,0)$, $G \mapsto (1,1)$. A codon $b_1b_2b_3$ is then represented as the concatenation $\varphi(b_1) \parallel \varphi(b_2) \parallel \varphi(b_3) \in \mathrm{GF}(2)^6$. The 64 codons become the 64 vertices of the 6-dimensional hypercube Q_6. This default bijection was adopted in companion methodological work (Clayworth (2026)) prior to the encoding-sensitivity analysis reported here, on the grounds of visualization clarity: it places the standard code's nine 2-fold-degenerate amino acids on bit-5 differences (the natural geometry of wobble degeneracy). We do not claim this encoding is biologically privileged. All encoding-dependent results are reported under all 24 base-to-bit bijections (Section 2.1, end), and the load-bearing topology-avoidance result uses the encoding-independent $H(3,4)$ adjacency (Section 2.3.4).

Under this encoding, two codons that differ by a single-nucleotide substitution in which only one bit of the 2-bit pair changes are adjacent in Q_6 (Hamming distance 1). However, transversions that flip both bits within a nucleotide position correspond to Hamming distance 2. The full single-nucleotide mutation graph is therefore the Hamming graph $H(3,4) = K_4 \,\square\, K_4 \,\square\, K_4$ (the Cartesian product of three complete graphs on the four nucleotide states; equivalently, $K_4^{\square 3}$), with 64 vertices, regular degree 9, and 288 undirected edges. It contains Q_6 as a 192-edge subgraph: Q_6 contributes the 192 Hamming-distance-1 edges (single-bit changes within one 2-bit nucleotide), while the remaining 96 within-nucleotide diagonal (Hamming-distance-2) edges complete $H(3,4)$. To address the concern that Q_6 misses approximately one-third of single-nucleotide mutations, we introduce the weighted score F_ρ that interpolates between pure Q_6 ($\rho = 0$) and full $H(3,4)$ ($\rho = 1$); see Section 2.3. Throughout this paper $H(3,4)$ denotes this nucleotide-level mutation graph, which is encoding-independent (every two-bit bijection from $\{A,C,G,U\}$ to $\{0,1\}^2$ yields the same $H(3,4)$); Q_6 is encoding-dependent, since the partition into Hamming-1 vs Hamming-2 edges depends on the bijection.

There are 24 distinct bijections from $\{C,U,A,G\}$ to $\mathrm{GF}(2)^2$ (all permutations of $4! = 24$ assignments). All encoding-dependent results are tested across all 24 encodings; encoding-invariant

properties (such as Serine's disconnection at $\varepsilon = 1$, where ε denotes the Hamming distance threshold in $\mathrm{GF}(2)^6$) are noted as such. Coloring optimality is significant ($p < 0.05$) under all 24 encodings; full encoding-sensitivity results are provided in the Supplementary Material.

2.2 Edge-mismatch objective function

The genetic code assigns each vertex $v \in Q_6$ a label $c(v) \in \mathcal{A}$, where $\mathcal{A}$ comprises the 20 amino acids and the stop signal. The edge-mismatch score is

$$F(c) = \sum_{\{v,w\}:d(v,w)=1} \Delta(c(v), c(w)) \tag{1}$$

where the sum ranges over all 192 edges of Q_6 (pairs of vertices at Hamming distance 1), and Δ is a physicochemical distance between amino acids. We test four distinct distance metrics: the Grantham (1974) composite distance (composition, polarity, volume; range 5–215), the Miyata et al. (1979) normalized Euclidean distance (polarity and volume only; range 0.06–5.13), the Woese polar requirement absolute difference (range 0–8.2; (Woese, 1973; Woese et al., 1966); used by Freeland & Hurst (1998)), and the Kyte & Doolittle (1982) hydropathy absolute difference (range 0–9.0; used by Haig & Hurst (1991)). Synonymous edges contribute 0; edges involving a stop codon receive a fixed penalty scaled proportionally to each metric's maximum. A lower F indicates a more error-minimizing code.

Stop codons are held fixed across all null models (Section 2.3) because their assignment is constrained by release-factor recognition geometry rather than tRNA decoding: eRF1 binds UAR/UGA stop codons via a structurally distinct mechanism from sense-codon decoding, and permuting stops would conflate two evolutionarily separate optimization problems. The stop-codon contribution is therefore a constant offset; sensitivity analysis across penalty values (0, 150, 215, 300) confirms this is immaterial to the ranking.

2.3 Null models

2.3.1 Block-preserving null (Freeland–Hurst)

Following Freeland & Hurst (1998), we group the 64 codons into 16 blocks of 4, defined by shared first-two-base prefix. Each block's internal pattern of amino acid assignments is preserved, but the mapping of patterns to blocks is permuted uniformly at random. Blocks containing stop codons are held fixed. This null preserves the synonymous codon contiguity (wobble degeneracy) inherent in the genetic code, providing a stringent test: the observed code must beat random codes that share its degeneracy architecture.

For the standard code, we drew $n =$10,000 null samples (seed 135325) and computed the conservative p-value as $\frac{k+1}{n+1}$, where k is the number of null scores below the observed F.

2.3.2 Per-table null

The NCBI Genetic Codes registry (https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi; gc.prt v4.6, retrieved 2026-04-25) currently lists 27 translation tables: codes 1–6, 9–16, and 21–33 (codes 7, 8, and 17–20 are deprecated and absent from the current registry). All 27 are analyzed in this work. Two pairs of tables share identical sense-codon mappings and differ only in their start-codon assignments: tables 1 (Standard) and 11 (Bacterial / Archaeal / Plant Plastid), and tables 27 (Karyorelict Nuclear) and 28 (Condylostoma Nuclear). Both pairs are retained as separate entries to match NCBI numbering, but produce identical results for analyses that depend only on codon→amino-

acid mappings; the 27 NCBI tables thus correspond to 25 distinct sense-codon colorings. Each table was tested against its own block-preserving null (n =10,000 per table, common-seed design with seeds = base seed + table ID). P-values were corrected for multiple comparisons using the Benjamini & Hochberg (1995) false discovery rate (FDR) procedure. For NCBI tables with dual-function stop/sense codons (e.g., table 27 lists UGA as "Stop or Trp"; table 28 lists UAA, UAG as "Gln or Stop" and UGA as "Trp or Stop"), the primary codon→amino-acid analyses use the amino-acid label given in the NCBI AAs row of the gc.prt definition; stop functionality is handled only in analyses explicitly involving stop labels.

2.3.3 Graph family G_ρ and rho sweep

We define a family of mutation graphs G_ρ parameterized by $\rho \in [0, 1]$, where $G_0 = Q_6$ (192 Hamming-1 edges) and $G_1 = H(3, 4)$ (all 288 single-nucleotide substitution edges). The weighted mismatch score on G_ρ is

$$F_{\rho(c)} = F_{\text{Hamming-1}}(c) + \rho \cdot F_{\text{diagonal}}(c) \tag{2}$$

where F_{diagonal} sums over the 96 within-nucleotide distance-2 edges. This formulation treats Q_6 not as the biological object but as one endpoint of a continuous interpolation toward the full mutation graph. We evaluated five values of $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ with n =10,000 block-preserving null samples per value (common-seed design). Conservative p-values, computed as $\frac{k+1}{n+1}$ where k is the number of null samples scoring below the observed code: $p_{\rho=0} = 0.0061$ ($k = 60$), $p_{\rho=0.25} = 0.0023$ ($k = 22$), $p_{\rho=0.5} = 0.0007$ ($k = 6$), $p_{\rho=0.75} = 0.0003$ ($k = 2$), $p_{\rho=1} = 0.0003$ ($k = 2$). The last two are near the Monte Carlo resolution limit (the absolute lower bound at n =10,000 is $\frac{1}{n+1} \approx 1 \times 10^{-4}$, attained only when $k = 0$); enlarging n would not change the qualitative conclusion that the observed code lies in the extreme low-cost tail at every ρ.

2.3.4 Table-preserving permutation null (topology avoidance)

To test whether natural codon reassignment events avoid disrupting amino acid connectivity, we define a single primary candidate universe of single-codon reassignments and treat all reasonable alternatives as a sensitivity analysis. From the standard code C, the primary candidate set $\mathcal{M}(C)$ pairs each of the 64 codons with each of the 20 alternative labels drawn from $\mathcal{A}_{20} \cup \{\text{Stop}\}$:

$$\mathcal{M}(C) = \{(x, y) : x \in \text{codons}, y \in \mathcal{A}_{20} \cup \{\text{Stop}\}, y \neq C(x)\}, \quad |\mathcal{M}(C)| = 64 \times 20 = 1280. \tag{3}$$

This **21-label, identity-excluded** universe (denoted U1 in Supplement §S5) gives every codon exactly 20 alternative labels and admits stop-codon reassignment, which is biologically attested. Identity moves ($y = C(x)$) are excluded because they contribute no signal. Two strict variants—the **amino-acid-only, identity-excluded** universe with $|\mathcal{M}| = 61 \times 19 + 3 \times 20$ =1,219 (U2) and the **stop-inclusive with no-ops** universe with $|\mathcal{M}| = 64 \times 21$ =1,344 (U4)—are reported as sensitivity analyses in Supplement §S5; topology-avoidance results are qualitatively identical under all three.

For each amino acid a, let G_a^1 denote the subgraph induced on codons assigned to a with edges between codons at binary Hamming distance ≤ 1. A reassignment is *topology-breaking* if it increases the number of connected components of G_a^1 for any amino acid a (the **increase-in-components** definition $\Delta\beta_0 > 0$ used by the conditional logit; the **new-disconnection-in-previously-connected-family** definition is reported in parallel as a sensitivity check; see Supplement §S3). From the 27 NCBI translation tables analyzed in this work (Section 2.3.2), we catalogued the table-specific reassignment events relative to the standard code; de-duplicating by (codon, target amino acid) tuple yields the unique-event set on which the hypergeometric and table-preserving permutation tests operate. The

conditional logit model (Section 2.3.5) operates on the full table-specific event-step list, which preserves recurrent reassignments across independent lineages.

Two tests were applied: (i) a hypergeometric test treating the observed events as a sample from the finite candidate landscape (N =1,280, $K = 846$ topology-breaking under the primary cell—$H(3,4)$ adjacency with $\Delta\beta_0 > 0$ definition—against $n = 28$ observed, $x = 6$ topology-breaking observed); and (ii) a table-preserving permutation test (n =10,000, seed 135325) that permutes target amino acids among a table's reassigned codons, preserving within-table codon and target structure to address phylogenetic non-independence. Risk ratios with 95% log-normal confidence intervals were computed as $\mathrm{RR} = (x/n)/(K/N)$. The full 2×2 definition $\times$ adjacency audit (Q_6 vs $H(3,4)$ adjacency $\times$ new-disconnection vs $\Delta\beta_0 > 0$ definitions) is given in Supplement §S3; risk ratios fall in 0.28–0.33 across all four cells.

2.3.5 Conditional logit model of reassignment choice

To test whether topology avoidance contributes independent explanatory power beyond physicochemical optimization, we fit event-level conditional logit (discrete-choice) models to natural reassignment events. Each observed reassignment is treated as a choice from the set of all ≈1,280 possible single-codon reassignments available at the current code state. For a table with k reassignment events, the code state evolves sequentially: at each step, the model assigns a probability to each candidate move based on a linear utility score, and the observed move is compared against all alternatives.

For each candidate move m from code state C, we computed three features: (i) Δ_{phys}, the change in local Grantham mismatch cost summed over Hamming-1 edges incident to the reassigned codon; (ii) Δ_{topo}, the total increase in connected components across all amino acid codon graphs at $\varepsilon = 1$; and (iii) Δ_{tRNA}, the Hamming distance from the reassigned codon to the nearest codon already encoding the target amino acid (a heuristic proxy for tRNA repertoire disruption). The conditional logit probability of observing move m^* is

$$P(m^* \mid \mathcal{N}(C)) = \frac{\exp(\boldsymbol{w}^\top \boldsymbol{x}_{m^*})}{\sum_{m \in \mathcal{N}(C)} \exp(\boldsymbol{w}^\top \boldsymbol{x}_m)} \tag{4}$$

where $\boldsymbol{w}$ is the weight vector and $\boldsymbol{x}_m$ is the feature vector for move m. Features were z-scored across all candidates for numerical stability.

Since the temporal ordering of reassignment events within a table is unknown, we marginalized the likelihood over all $k!$ orderings for tables with $k > 1$ changes:

$$L_{\mathrm{table}} = \frac{1}{k!} \sum_{\sigma \in S_k} \prod_{s=1}^{k} P\Big(m^*_{\sigma(s)} \mid \mathcal{N}\big(C_{\sigma,s}\big)\Big) \tag{5}$$

For tables with $k \leq 6$ events (including the largest, yeast mitochondrial, $k = 6$), we enumerate all $k!$ orderings exactly (up to 720 for $k = 6$); for the rare cases of $k > 6$, we sample 720 random orderings with the seeded RNG. The total likelihood is the product across all 25 tables with reassignment events (66 total event-steps).

Four nested models were compared under Q_6 topology: M1 (physicochemistry only, $w_{\mathrm{topo}} = w_{\mathrm{tRNA}} = 0$), M2 (topology only, $w_{\mathrm{phys}} = w_{\mathrm{tRNA}} = 0$), M3 (physicochemistry + Q_6 topology, $w_{\mathrm{tRNA}} = 0$), and M4 (all three features). Two additional verification variants were fit using the encoding-independent $H(3,4)$ topology feature in place of Q_6: $\mathrm{M2}_{\mathrm{H(3,4)}}$ (topology only, $H(3,4)$) and $\mathrm{M3}_{\mathrm{H(3,4)}}$ (physicochemistry + $H(3,4)$ topology). The $H(3,4)$ variants test whether M3 dominance is robust to the choice of topology graph; the $\Delta_{\mathrm{topo,H(3,4)}} = \sum_a \Big(\beta_0\big(G_a^{\mathrm{after,H(3,4)}}\big) - \beta_0\big(G_a^{\mathrm{before,H(3,4)}}\big)\Big)$

feature replaces the Q_6 component-count change. Weights were estimated by maximum likelihood (scipy.optimize, Nelder–Mead with L-BFGS-B refinement). Model comparison used the corrected Akaike Information Criterion (AICc) and likelihood-ratio tests for nested pairs.

2.3.6 Fisher–Stouffer test for tRNA enrichment

For each variant-code organism paired with a phylogenetically proximate standard-code control, we built a 2×2 contingency table comparing the proportion of tRNA genes encoding the reassigned amino acid versus all other amino acids. The sampling model treats tRNA gene counts as draws from a hypergeometric distribution conditional on the row and column marginals (total tRNAs per organism, total tRNAs for the focal amino acid), which is appropriate when the question is whether a specific amino acid's share of the tRNA repertoire differs between two genomes. Fisher's exact test (one-sided, alternative = "greater") was applied per pairing, and p-values were combined via Stouffer's Z method. To address non-independence from shared control organisms, we constructed a conflict graph (edges connect pairings sharing an organism) and enumerated all maximal independent sets (MIS) via the Bron–Kerbosch algorithm with pivoting. For each MIS with ≥ 2 members, we computed the Stouffer combined p-value and report the worst-case (maximum p) across all MIS as the primary conservative test.

tRNA gene counts for 18 organisms were obtained by running tRNAscan-SE 2.0.12 (Chan & Lowe, 2019) with Infernal 1.1.4 on NCBI genome assemblies (eukaryotic mode for ciliates and yeast; bacterial mode for *Mycoplasma*). The verified set comprises 5 organisms with translation table 6 (UAA/UAG→Gln; ciliates: *Tetrahymena thermophila*, *Paramecium tetraurelia*, *Oxytricha trifallax*, *Pseudocohnilembus persalinus*, *Halteria grandinella*), 6 with table 10 (UGA→Cys; *Euplotes* species), 1 with table 15 (*Blepharisma stoltei*; the genome tested here reads UGA→Trp via a dedicated suppressor tRNA-Trp(UCA) (Singh et al., 2023), despite the legacy NCBI table-15 UAG→Gln assignment, see Table 8 footnote §), 1 with table 31 (multiple stops reassigned; *Blastocrithidia nonstop*), 2 with table 4 (UGA →Trp; *Mycoplasmoides genitalium*, *M. pneumoniae*), and 3 standard-code controls (*Stentor coeruleus*, *Ichthyophthirius multifiliis*, *Fabrea salina*). One additional standard-code organism (*Saccharomyces cerevisiae*) was sourced from GtRNAdb (https://gtrnadb.ucsc.edu) for context but is not included in the primary 24-pairing analysis. The dataset comprises 24 pairings across 5 variant genetic codes (NCBI translation tables 4, 6, 10, 15, and 31) and 3 tRNAscan-SE–verified standard-code controls.

2.4 Retrospective cross-study reanalysis of synthetic recoding outcomes

To test whether GF(2)^6 topology predicts codon recoding outcomes in synthetic biology experiments, we performed a retrospective cross-study reanalysis across nine published genome recoding datasets, with quantitative analysis of eight (the ninth, (Ding et al., 2024), is included for cross-kingdom scope without quantitative extraction). For each codon substitution reported in these studies, we computed three topology features under the default encoding ($C \mapsto (0,0)$, $U \mapsto (0,1)$, $A \mapsto (1,0)$, $G \mapsto (1,1)$):

1. **Boundary crossing** ($\varepsilon = 1$): whether the source and target codons lie in different connected components of the amino acid's codon graph at Hamming distance 1. Generalized beyond serine to all amino acids with potential disconnections (including leucine, arginine).
2. **Local mismatch change** (ΔF_{local}): the difference in total Grantham distance summed over all Hamming-1 neighbors between the target and source codon positions: $\Delta F_{\text{local}} = \sum_{n \in N(t)} \Delta(c(t), c(n)) - \sum_{n \in N(s)} \Delta(c(s), c(n))$, where $N(v)$ denotes the 6 Hamming-1 neighbors of vertex v in Q_6.

3. **Hamming distance**: the number of bit positions differing between source and target vectors in GF(2)^6.

Codon positions were extracted from published GenBank genome files by CDS-level comparison using Biopython (Cock et al., 2009), with codon frame offsets (`/codon_start`) handled explicitly. CDS features were matched between parent and recoded genomes by locus tag. The primary analysis used the Syn57 dataset (Robertson et al., 2025), which provides a within-study contrast: 37,146 serine recodings cross the UCN↔AGY family boundary, while 22,859 alanine recodings remain within the GCN family. Syn57 RNA-seq differential expression data (Data file S10) were used as the outcome measure. Design-to-final genome deviations were identified by comparing the vS33A7 design genome (Data file S2) against the verified Syn57 genome (Data file S8), filtering genes with >10 CDS-level differences as structural rearrangements (sensitivity analysis across thresholds 3–50 reported in the Supplementary Material). The Ostrov et al. (2016) segment viability data (Table S4) were analyzed as a case-control comparing segments with and without lethal design exceptions; Bonferroni correction was applied for three simultaneous tests.

All cross-study reanalysis code is released in the `codon_topo.analysis.codonsafe` subpackage of the `codontopo` repository (https://github.com/biostochastics/codontopo, version 0.4.0) and can be run via `codon-topo codonsafe`. Raw data files and download provenance are documented in `data/codonsafe/DATA_MANIFEST.md`.

2.5 Synthetic-biology feasibility score (visualization-only)

For Figure 4A we use a heuristic feasibility score $S(m) \in [0, 1]$ that combines four-fold filtration preservation, a smooth function of local Grantham mismatch change, and Hamming accessibility to the target amino acid. The score is purely a visualization aid for delineating high- versus low-feasibility regions of the candidate single-codon reassignment landscape; it is not used in any inferential test in this paper. The exact functional form, weights, and implementation are given in Supplement §S12 and `src/codon_topo/analysis/synbio_feasibility.py` of the public repository.

2.6 Claim hierarchy

All 15 scientific claims evaluated in this work are registered in a formal claim hierarchy with pre-specified status categories (Table 1): SUPPORTED (passes rigorous null, $p < 0.01$), SUGGESTIVE (trend-level, $p < 0.05$), EXPLORATORY (hypothesis-generating), REJECTED (falsified or pre-rejected in literature), FALSIFIED (directly contradicted by data), and TAUTOLOGICAL (true by construction). The full hierarchy with justifications is available in the supplementary materials.

3 Results

3.1 Cross-metric coloring optimality

Under the Freeland–Hurst block-preserving null with $n = 10{,}000$ permutations, the standard genetic code is significantly low-cost across all four physicochemical distance metrics examined (Table 2). For Grantham distance, the observed score is $F = 13{,}477$ versus a null mean of $14{,}954 \pm 628$ ($z = 2.35$, quantile 0.6%, $p = 0.006$). The same pattern holds for Miyata distance ($F = 235.3$ vs 275.7 ± 12.9, $z = 3.14$, $p < 0.001$), Woese polar requirement ($F = 347.3$ vs 410.5 ± 23.1, $z = 2.73$, $p = 0.003$), and Kyte–Doolittle hydropathy ($F = 457.2$ vs 570.7 ± 40.9, $z = 2.78$, $p = 0.001$). Relative to null expectations, the observed code lowers the mismatch score by 9.9%, 14.6%, 15.4%,

Status	Claim	Null model	p-value
Supported	Cross-metric coloring optimality	FH block-preserving $\times$ 4 metrics	≤ 0.006
	Per-table optimality (26/27)	FH per-table + BH-FDR	< 0.05
	ρ-robustness ($\rho \in [0, 1]$)	FH weighted edges	0.006
	Topology avoidance (Q_6 + $H(3, 4)$)	Table-pres. perm. + clade excl.	10^{-4}
Suggestive	tRNA enrichment (MIS worst-case)	Fisher–Stouffer (MIS)	0.045
Exploratory	Bit-position bias (dedup.)	χ^2 uniform	0.075
	Mechanism boundary conditions	Descriptive	—
	Atchley F3/Serine convergence	Qualitative	—
	Disconnection catalogue (4 cases)	Descriptive	—
Falsified	KRAS–Fano clinical prediction	Fisher–Bonferroni	1.0
Rejected	Serine min-distance-4 invariant	Counterexample	—
	PSL(2,7) symmetry	Literature	—
	Holomorphic embedding	Verification	—
Tautological	Two-fold bit-5 filtration	—	—
	Four-fold prefix filtration	—	—

Table 1: Summary of 15 evaluated claims. "FH" = Freeland–Hurst. "MIS" = maximal independent set enumeration. p-values are the most conservative test per claim. Shaded rows: claims that failed validation.

and 19.9%, respectively. Beta-posterior credible intervals for the empirical p-values remain entirely below 0.01 for all four metrics (computed from the conservative $\frac{k+1}{n+1}$ estimator with $\mathrm{Beta}(k+1, n-k+1)$ posterior; per-metric intervals are listed in `output/coloring_optimality.json` under `beta_credible_intervals`). Since Freeland & Hurst (1998) established optimality using polar requirement alone, the cross-metric concordance demonstrates that error-minimization is robust across multiple distinct physicochemical parameterizations, not an artifact of any single distance metric. Under a weaker degeneracy-only null that preserves only codon family sizes without maintaining block contiguity, the signal strengthens substantially ($z > 9, p < 10^{-4}$ for all metrics), confirming that the block-preserving null provides a stringent test.

Decomposing F by nucleotide position reveals that the second codon position contributes 49.3% of the total mismatch, followed by the first position at 38.2% and the wobble position at 12.5%. This gradient reflects the well-known biochemical hierarchy: second-position mutations are the most physicochemically disruptive, first-position mutations are intermediate, and wobble-position mutations are largely synonymous.

3.2 Robustness across diagonal-edge weight ρ

The weighted score F_ρ (Equation 2) remains significantly optimal across all values of ρ from 0 to 1 (Table 3). At $\rho = 0$ (pure Q_6), $p = 0.006$ ($z = 2.35$); at $\rho = 1$ (full $H(3, 4)$, all 288 single-nucleotide edges equally weighted), $p < 0.001$ ($z = 3.37$). No value of ρ yields $p > 0.006$. The optimality signal strengthens monotonically as diagonal edges are included, addressing the concern that the Q_6 representation ignores approximately one-third of single-nucleotide mutations.

Metric	Observed	Null mean ± SD	z	p	Improv.
Grantham (Grantham, 1974)	13,477	14,954 ± 628	2.35	0.006	9.9%
Miyata (Miyata et al., 1979)	235.3	275.7 ± 12.9	3.14	< 0.001	14.6%
Polar requirement (Woese, 1973)	347.3	410.5 ± 23.1	2.73	0.003	15.4%
Kyte–Doolittle (Kyte & Doolittle, 1982)	457.2	570.7 ± 40.9	2.78	0.001	19.9%

Table 2: Cross-metric sensitivity analysis. Block-preserving null ($n = 10{,}000$, seed 135325) under four distinct physicochemical distance metrics. All metrics show significant optimality ($p < 0.01$). Effect size $z = (\mu_{\text{null}} - F_{\text{obs}})/\sigma_{\text{null}}$. "Improv." = percent improvement of observed score over null mean. Under a weaker degeneracy-only null (preserving only codon family sizes, not block structure), all four metrics yield $z > 9$ and $p < 10^{-4}$, confirming the result is not an artifact of the Freeland–Hurst (Freeland & Hurst, 1998) block-preserving design.

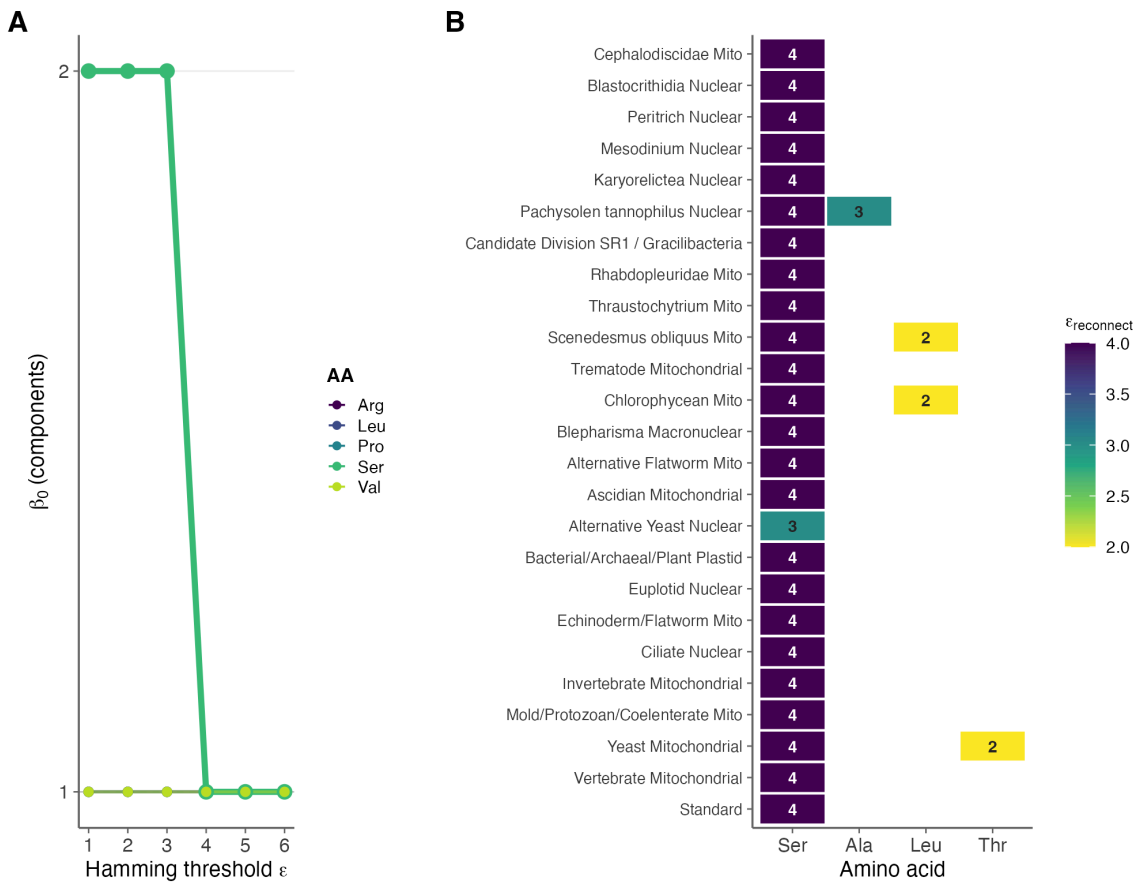

Figure 1: Core topology of the genetic code in $\mathrm{GF}(2)^6$. **(A)** Persistent homology: connected components (β_0) of amino acid codon graphs as a function of Hamming distance threshold ε. Serine (bold) is the only amino acid disconnected at $\varepsilon = 1$ across all 27 NCBI translation tables, reconnecting at $\varepsilon = 4$. **(B)** Disconnection catalogue across all translation tables. Each tile denotes an amino acid disconnected at $\varepsilon = 1$; the number is the reconnection ε. Serine is universally disconnected; Leu, Thr, and Ala appear as variant-code disconnections.

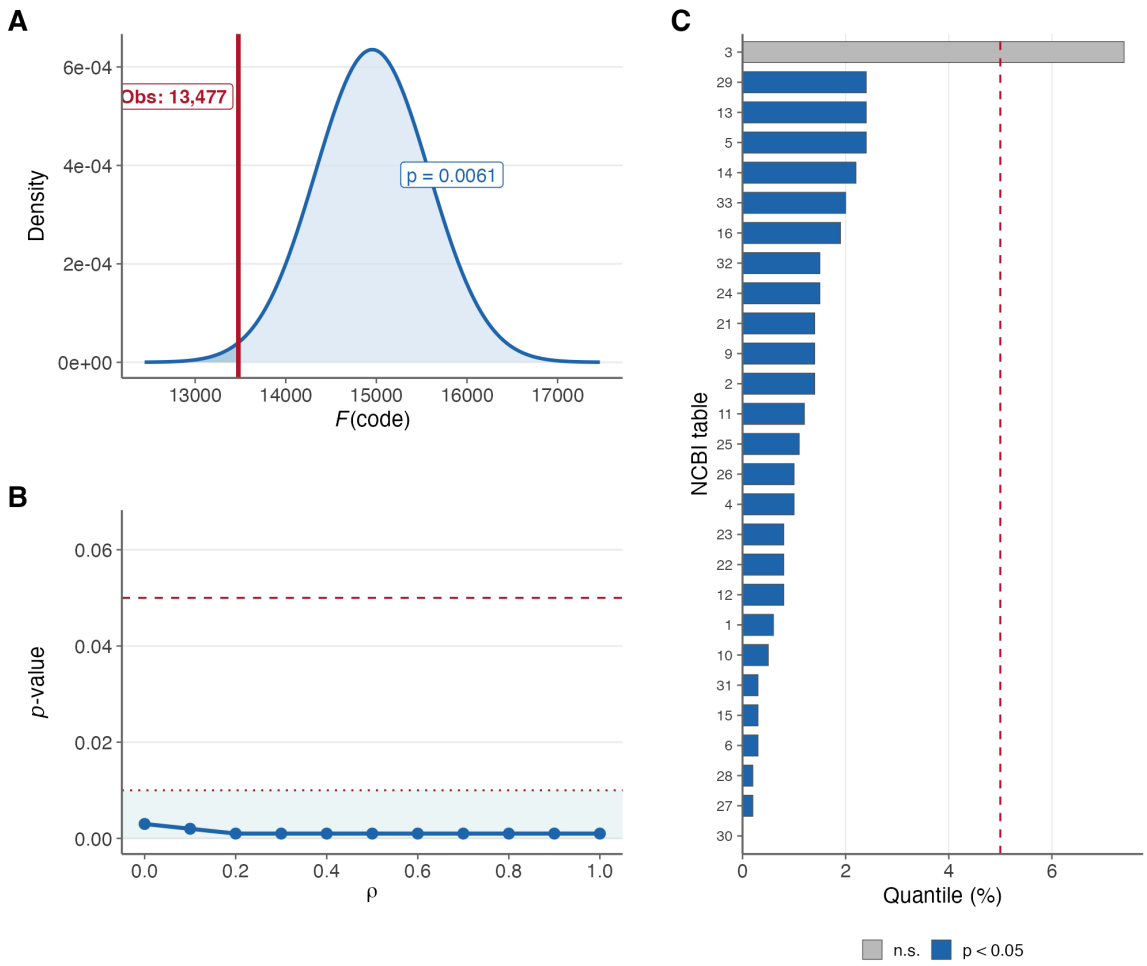

Figure 2: Coloring optimality of the genetic code. **(A)** Null distribution of Grantham edge-mismatch scores F under the Freeland–Hurst block-preserving null ($n = 10{,}000$). The observed standard code (red line, $F = 13{,}477$) falls at the 0.6th percentile ($p = 0.006$). **(B)** p-value across diagonal-edge weight ρ from 0 (Q_6) to 1 (full $H(3,4)$); all values below 0.01, with optimality strengthening monotonically. **(C)** Per-table quantile for each of 27 NCBI translation tables under their own block-preserving null (n =10,000); only table 3 (yeast mito.) exceeds the 5% threshold.

ρ	Observed	Null mean	Quantile	p-value	z
0	13,477	14,954	0.6%	0.006	2.35
0.25	15,335	17,020	0.22%	0.002	2.77
0.5	17,193	19,085	0.06%	< 0.001	3.09
0.75	19,050	21,150	0.02%	< 0.001	3.28
1	20,908	23,216	0.02%	< 0.001	3.37

Table 3: Robustness of coloring optimality across diagonal-edge weight ρ. Block-preserving null, $n = 10{,}000$ per ρ. All p-values < 0.01. Quantiles reported as empirical rank; values $\leq 0.02\%$ indicate ≤ 2 null samples below the observed score.

3.3 Per-table optimality preservation

Of the 27 NCBI translation tables, 26 remain in the top 5% of their own block-preserving null after Benjamini–Hochberg FDR correction (Figure 2, panel C). The mean quantile across all tables is 1.4%. The single marginal exception is translation table 3 (yeast mitochondrial code, 6 codon reassignments; p_{BH} =0.075).

The headline number conflates two distinct evidentiary regimes, and we disaggregate them explicitly. Variant tables that differ from the standard code by ≥ 3 codon reassignments (henceforth "informative-distance" tables) sit far enough from the standard code that the per-table block-preserving null contains many permutations that are not close to the standard code, so the test genuinely probes whether the variant is independently low-cost. Variant tables with ≤ 2 reassignments (henceforth "near-standard" tables) sit so close to the standard code that the per-table null distribution is dominated by permutations near the standard code, and the test cannot reliably distinguish "this variant is independently optimal" from "this variant is close to the standard code, which is optimal." A standard-code-proximity audit (Supplement §S8) confirms this disaggregation empirically: for informative-distance tables the conditional-on-distance quantile agrees with the unconditional quantile; for near-standard tables it does not.

Of the 12 informative-distance tables, 11 retain top-5% placement after BH–FDR correction (the lone marginal case is yeast mitochondrial). Of the 15 near-standard tables, 15 are formally significant under the same correction, but those results should be read as **confirmatory of standard-code geometry** rather than as evidence for independent variant-specific optimization. The substantive evolutionary claim is therefore: of the variant codes that differ enough from the standard code for the test to be informative, all but one (the most extensively reassigned, yeast mitochondrial) preserve near-optimal physicochemical placement. Even the most reassigned code (table 3, 6 codons reassigned) sits at the 7.4th percentile of its block-preserving null, marginally above the 5% threshold but still in the lower tail.

3.4 Topology avoidance in natural codon reassignments

We adopt the encoding-independent $H(3,4)$ Hamming graph as the primary adjacency for the topology-avoidance test, with the Q_6 subgraph reported as a coordinate-dependent decomposition (motivated below). Under $H(3,4)$, of the 1,280 candidate single-codon relabelings (Methods Methods §2.3.5), 846 (66.1%) increase the connected-component count of some amino acid's codon graph relative to the standard code ($\Delta\beta_0 > 0$). Only 6 of 28 observed natural reassignment events (21.4%) do so, yielding a 3.1-fold depletion (RR $= 0.32$, 95% CI $[0.16, 0.66]$; hypergeometric $p =$ 0.0000012798706572815002; table-preserving permutation $p \leq 10^{-4}$).

Under Q_6 (Hamming-1 adjacency in the default GF$(2)^6$ encoding), the corresponding numbers are 931 of 1,280 candidates (72.7%) creating new amino-acid disconnections, against only 6 of 28 observed events (21.4%); Q_6 depletion is 3.4-fold (hypergeometric $p = 0.000000015839628551544103$; permutation $p \leq 10^{-4}$). The Q_6 candidate-rate is somewhat higher than the H(3,4) rate because some Q_6-disconnected pairs become connected when within-nucleotide Hamming-2 edges are admitted. Both adjacencies yield highly significant depletion under both topology-breaking definitions (new disconnection in a previously connected family; $\Delta\beta_0 > 0$ increase in components), with risk ratios in the range 0.28–0.33 across the four cells of the 2×2 definition $\times$ adjacency audit (Supplement §S3). The avoidance of topology-disrupting reassignment trajectories is therefore not specific to either the hypercube subgraph or to either definition. We give precedence to the $H(3,4)$ result because Q_6 is encoding-dependent: across all 24 base-to-bit bijections, the Q_6 candidate-landscape rate ranges from

36% (8 encodings) to 73% (default encoding), with median hypergeometric $p = 6 \times 10^{-6}$ but $p > 0.5$ in 8 of 24 encodings (Supplement §S4). $H(3,4)$, by contrast, depends only on nucleotide identity and is robust by construction.

Following the phylogenetic-distribution analysis of mitochondrial reassignment events by Sengupta et al. (2007), we performed clade-exclusion sensitivity analysis, iteratively removing each major taxonomic group (all ciliates, all metazoan mitochondria, all CUG-clade yeasts, etc.) and retesting. The depletion is highly significant in every regime ($p < 10^{-5}$). However, the structure of this finding warrants honest reading: the topology-breaking events themselves are not pan-taxonomic. Of the 6 topology-breaking reassignments under $H(3,4)$, 4 are concentrated in yeast mitochondrial (table 3), and excluding that single clade **strengthens** the depletion (observed rate drops from 21.4% to 8.3%; hypergeometric p from 1.58×10^{-8} to 3.6×10^{-11}). The clade-exclusion robustness is therefore a denominator effect—removing clades that contributed zero or one topology-breaking event each leaves the breakage rate essentially unchanged—rather than a numerator effect that would demonstrate breaking-then-failing across multiple lineages. The honest reading is: the **avoidance** of topology-breaking moves is pan-taxonomic, supported by 22 of 28 events that did not break topology distributed across many clades; the **attempts** to break topology are concentrated in one mitochondrial code, and with only 6 breakers the cross-clade structure of the breaker set is genuinely underpowered.

This concentration is itself informative. Yeast mitochondrial is also the marginal outlier in the per-table optimality analysis (§3.3), the conditional-logit case with the lowest observed-move percentile (§3.5), and the only variant code requiring acquisition of a novel tRNA derived from a different parent (tRNA^{Thr} from tRNA^{His} (Su et al., 2011)). Under the null hypothesis that table 3 is just one of 27 tables drawn at random, the prior probability that one specific table converges across three independent analyses is on the order of $\frac{1}{27^2} \approx \frac{1}{700}$, so this convergence is supporting evidence rather than coincidence. We treat it as evidence that yeast mitochondrial occupies a distinct position in the codon-reassignment landscape—one where multiple constraints (optimality, topology, mechanism) line up in the same direction—rather than as evidence for or against any specific selectionist mechanism.

The topology-breaking definition matters less than one might fear. The two natural definitions —**new disconnection in a previously connected family** (a candidate move makes some amino acid disconnected at $\varepsilon = 1$ that was connected in the standard code) and **increase in components** ($\Delta\beta_0 > 0$, the conditional-logit feature)—give qualitatively identical results across the $Q_6 \times H(3,4)$ adjacency $\times$ definition 2×2 matrix: 931, 963, 822, and 846 candidates of 1,280 are topology-breaking under the four definitions, with 5–7 of 28 observed events depending on cell, and all four cells yield hypergeometric $p < 10^{-5}$ with risk ratios in the range 0.28–0.33 (Supplement §S3). The encoding-sensitivity audit is more substantive. Across all 24 base-to-bit bijections, the encoding-independent $H(3,4)$ result is constant (these 24 encodings yield the same $H(3,4)$ graph), but the Q_6 subgraph and therefore the Q_6 candidate landscape varies substantially. Eight of the 24 encodings place the Q_6 candidate-landscape rate at approximately 36% (rather than 73% for the default encoding), and under those encodings the observed-rate of 21–36% does not differ from the candidate rate (median $p = 6 \times 10^{-6}$ across all 24 encodings, but 8 of 24 give $p > 0.5$). We therefore present $H(3,4)$ as the primary topology-avoidance result and treat Q_6 as a coordinate-dependent decomposition (Supplement §S4). Denominator sensitivity (alternative candidate universes: amino-acid-only $|M| = 1219$, stop-inclusive with no-ops $|M| = 1344$) is reported in Supplement §S5 and does not change the qualitative conclusion.

Quantity	Observed	Candidate landscape	Expected (chance)
Topology-breaking ($\Delta\beta_0 > 0$)	6 (21.4%)	846 (66.1%)	18.5 (66.1%)
Topology-preserving ($\Delta\beta_0 = 0$)	22 (78.6%)	434 (33.9%)	9.5 (33.9%)
Total	28	1,280	28
Depletion fold (cand. rate / obs. rate)	—	3.1×	1.0×
RR (95% CI)	—	0.32 (0.16–0.66)	1.00 (ref.)
Hypergeometric p (one-sided)	—	1.28×10^{-6}	(null reference)
Permutation p (table-pres.)	—	$\leq 10^{-4}$	(null reference)

Table 4: Topology avoidance in natural codon reassignments under the encoding-independent $H(3,4)$ Hamming graph (primary cell). Topology-breaking = $\Delta\beta_0 > 0$, an increase in the total number of connected components summed across amino acid codon graphs. The **Expected (chance)** column shows what would be observed under the null hypothesis that the 28 observed events are drawn uniformly at random from the 1,280-move candidate landscape: $E[\text{breaks}] = 28 \times K/N \approx 18.5$ versus 6 observed. RR = risk ratio (observed rate / candidate rate) with 95% log-normal CI; depletion fold = (candidate rate)/(observed rate). The remaining three cells of the 2×2 definition $\times$ adjacency audit (Q_6 vs $H(3,4)$ $\times$ new-disconnection vs $\Delta\beta_0 > 0$) are reported in Supplement §S3 (Table 5 of this manuscript reproduces them inline), with risk ratios in the range 0.28–0.33 and hypergeometric $p < 10^{-5}$ throughout.

3.5 Explanatory modeling: topology as an independent predictor

To test whether topology avoidance is reducible to physicochemical optimization, we fit event-level conditional logit models to the 66 reassignment event-steps across 25 variant-code tables, treating each observed reassignment as a choice among ≈1,280 candidate single-codon moves (Section 2.3.5). Six models were fit: four sharing the encoding-dependent Q_6 topology feature (M1 phys, M2 topo, M3 phys+topo, M4 phys+topo+tRNA) plus two H(3,4) verification variants that replace the $\Delta_{\text{topo,Q_6}}$ feature with the encoding-independent $\Delta_{\text{topo,H(3,4)}}$ ($\text{M2}_{\text{H(3,4)}}$ topo only, $\text{M3}_{\text{H(3,4)}}$ phys+topo). The combined physicochemistry-plus-topology model under Q_6 topology (M3) was strongly favored over all alternatives by AICc (AICc = 721.4), outperforming the physicochemistry-only model (M1; ΔAICc = 108.2) and the topology-only model (M2; ΔAICc = 89.1). Crucially, the encoding-independent verification model $\text{M3}_{\text{H(3,4)}}$ also strongly outperforms the physicochemistry-only baseline ($\Delta\text{AICc}(\text{M1} \rightarrow \text{M3}_{\text{H(3,4)}}) = 91.3$) and the H(3,4)-topology-only baseline ($\Delta\text{AICc}(\text{M2}_{\text{H(3,4)}} \rightarrow \text{M3}_{\text{H(3,4)}}) = 95.1$), so

Adjacency	Definition	K/N	Hyper. p	RR (95% CI)
$H(3,4)$ (primary)	$\Delta\beta_0 > 0$	846 / 1,280	1.3×10^{-6}	0.32 (0.16–0.66)
$H(3,4)$	new disconnection	822 / 1,280	5.0×10^{-7}	0.28 (0.13–0.62)
Q_6	$\Delta\beta_0 > 0$	963 / 1,280	2.2×10^{-8}	0.33 (0.17–0.63)
Q_6	new disconnection	931 / 1,280	1.6×10^{-8}	0.29 (0.14–0.60)

Table 5: Sensitivity analysis: the 2×2 topology-breaking definition $\times$ adjacency audit. All four cells share the same denominator (1,280 candidates) and the same observed-event total (28 de-duplicated reassignments); the observed topology-breaking count is 5–7 of 28 across cells. All four cells yield depletion in the same direction with comparable risk ratios (0.28–0.33). The $H(3,4)$ Hamming graph is encoding-independent; the Q_6 subgraph is encoding-dependent and 8 of 24 base-to-bit bijections give no Q_6 depletion at all (Supplement §S4). The main text uses the $H(3,4)$, $\Delta\beta_0 > 0$ cell as primary.

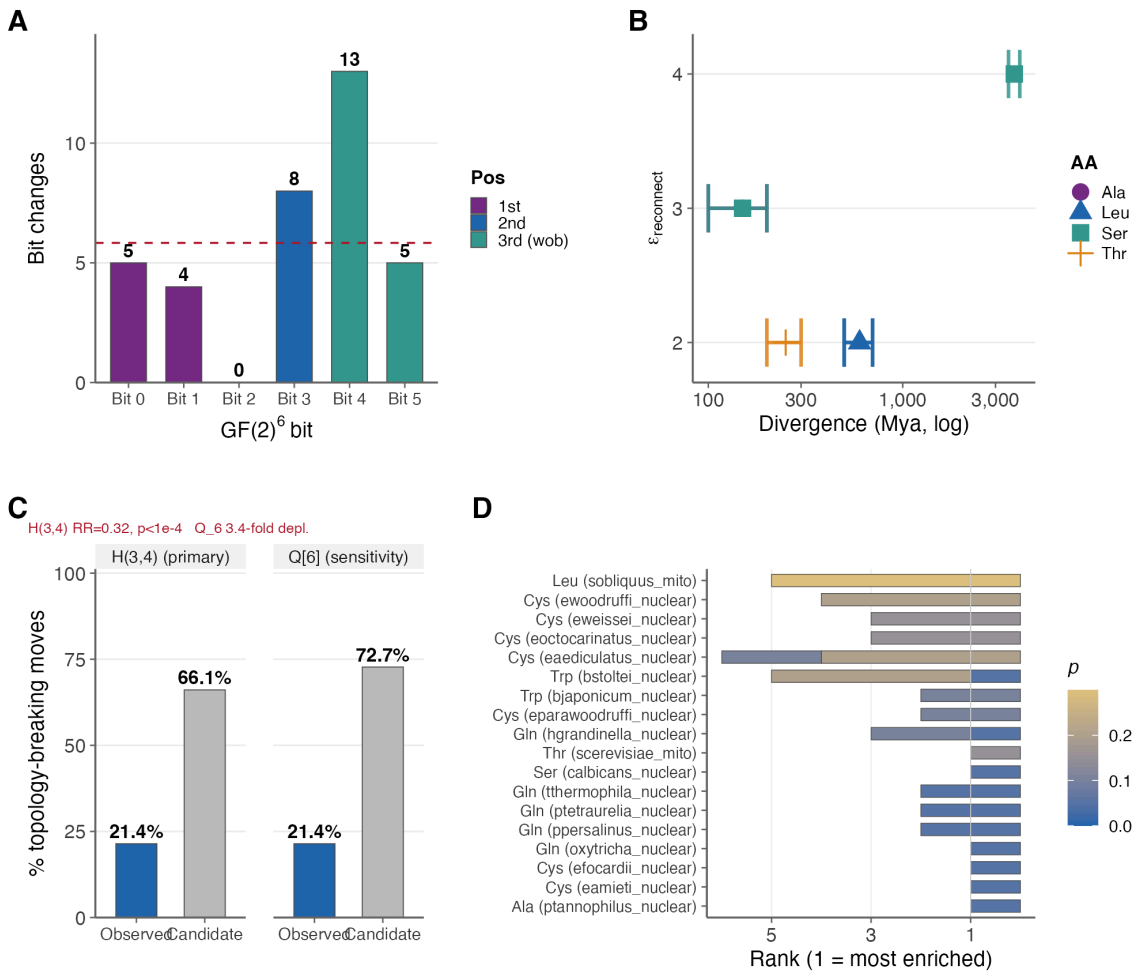

Figure 3: Evolutionary evidence for structural constraints on codon reassignment. **(A)** Bit-position bias: distribution of bit-flips across $\mathrm{GF}(2)^6$ coordinates in natural reassignment events; dashed line = uniform expectation. **(B)** Evolutionary depth calibration: reconnection ε vs estimated divergence age (log scale) for 4 variant-code amino acids. **(C)** Topology avoidance shown for the encoding-independent $H(3,4)$ Hamming graph (left panel, primary) and the encoding-dependent Q_6 subgraph (right panel, sensitivity). Under $H(3,4)$, observed natural reassignments break topology at 21.4% versus 66.1% of the candidate landscape (RR 0.32, permutation $p \leq 10^{-4}$). **(D)** tRNA enrichment: rank of the reassigned amino acid among all 20 AAs by tRNA gene proportion in variant-code vs standard-code organism pairings; rank 1 = most enriched.

the conditional-logit thesis does not depend on the choice of topology graph. Adding a heuristic tRNA-complexity proxy did not improve fit (M3→M4: $\Delta\mathrm{AICc} = 2.1$).

Likelihood-ratio tests confirmed that each feature class adds substantial explanatory value to the other: adding topology to physicochemistry yields $\mathrm{LR} = 110.4$ ($p \ll 10^{-10}$), and adding physicochemistry to topology yields $\mathrm{LR} = 91.2$ ($p \ll 10^{-10}$). The two feature classes are only weakly associated across the full candidate landscape, indicating limited confounding.

In the best-fitting model (M3), observed natural reassignments preferentially populate moves that reduce local physicochemical mismatch ($\hat{\beta}_{\mathrm{phys}} = -0.004$ per Grantham unit) and strongly avoid moves that increase amino acid codon-family disconnection ($\hat{\beta}_{\mathrm{topo}} = -3.26$ per additional connected component, where the conditional-logit feature uses the **increase-in-components** ($\Delta\beta_0 > 0$) defin-

Model	Description	k	Log $\mathcal{L}$	AICc	ΔAICc
M3	Phys + topo (Q_6, primary)	2	−358.6	**721.4**	**0**
M4	Phys + topo (Q_6) + tRNA proxy	3	−358.6	723.5	2.1
M3	Phys + topo (H(3,4) verification)	2	−367.1	738.4	16.9
M2	Topo only (Q_6)	1	−404.2	810.5	89.1
M1	Phys only	1	−413.8	829.7	108.2
M2	Topo only (H(3,4))	1	−415.7	833.5	112.1

Table 6: Event-level conditional logit model comparison. Six models share the same candidate set (≈ 1,280 single-codon moves per event-step) and the same likelihood structure marginalised over event orderings; they differ only in the feature subset used. The first four models use the encoding-dependent Q_6 topology feature (the legacy primary, paired with the encoding sweep in §S4); the two M-variants ending in $H(3,4)$ replace $\Delta_{\text{topo,Q_6}}$ with the encoding-independent $\Delta_{\text{topo,H(3,4)}}$ feature, providing the encoding-robustness verification. M3 (phys + topo) is favoured under both topology graphs; the gap between $\text{M3}_{\text{Q_6}}$ and $\text{M3}_{\text{H(3,4)}}$ reflects which topology feature carries more information per parameter, not whether topology is itself meaningful. k = number of estimated parameters. Bold = best model by AICc.

ition under Q_6 adjacency). Because Q_6 adjacency is encoding-dependent (8 of 24 base-to-bit bijections give no Q_6 topology depletion at the landscape level; Section 3.4), we additionally fit M3 with the encoding-independent $H(3,4)$ topology feature ($\Delta_{\text{topo,H(3,4)}}$ rather than $\Delta_{\text{topo,Q_6}}$). The encoding-robustness comparison gives $\Delta\text{AICc}(\text{M1} \to \text{M3}_{\text{H(3,4)}}) = 91.3$ and $\Delta\text{AICc}(\text{M2}_{\text{H(3,4)}} \to \text{M3}_{\text{H(3,4)}}) = 95.1$, both well above the conventional ΔAICc > 10 threshold treated as strong evidence in the model-comparison literature (Burnham & Anderson, 2002) (we note this threshold was empirically calibrated on linear-regression contexts and we therefore use it as a conventional reference rather than a formally calibrated cut-off in our setting), and similar in magnitude to the Q_6 counterparts ($\Delta\text{AICc}(\text{M1} \to \text{M3}_{\text{Q_6}}) = 108.2$). The M3 dominance is therefore not an artifact of the Q_6 encoding choice. Under M3, observed natural reassignments rank on average at the 89.5th percentile among all candidate moves (Figure 4, panel B), with the recurrent UGA→Trp reassignment consistently ranking above the 98th percentile. The one notable outlier is the yeast mitochondrial CUU→Thr reassignment (30th percentile), consistent with translation table 3 being the sole marginal exception in the per-table optimality analysis (Section 3.3). The non-significance of the heuristic tRNA-distance proxy (LR = 0.12, $p = 0.73$) speaks to the inadequacy of Hamming-distance-to-nearest-target-AA-codon as a proxy for tRNA-mediated mechanistic feasibility; it should not be interpreted as evidence against tRNA effects in general (see Section 3.6 for direct tRNA-gene-count tests). A posterior-predictive simulation under M3 reproduced the observed topology-breaking rate (observed 0.076 vs simulated mean 0.077; posterior-predictive $p = 0.60$), supporting model calibration rather than merely in-sample AICc improvement. Like all conditional logit models, M1–M4 assume Independence of Irrelevant Alternatives (IIA): the relative probability between any two candidate moves is unaffected by adding or removing other candidates. We use the model as an explanatory rather than predictive tool—the reported ΔAICc and likelihood-ratio statistics test whether topology adds explanatory value beyond physicochemical cost, not whether the model accurately predicts which specific reassignment will occur next; full IIA discussion appears in Supplement §S6. Conditional-logit clade-exclusion sensitivity (refitting M1–M4 with each major clade dropped, matching the regime applied to the topology-avoidance hypergeometric in Sengupta et al. (2007)) is reported in Supplement §S7. Across all seven exclusion regimes (ciliates, yeast mitochondrial, CUG-clade, metazoan mitochondrial, algal mitochondrial, protist mitochondrial, and hemichordate mitochondrial), M3 is robustly favored over M1 ($\Delta\text{AICc} \geq 49$, median 101, max 117) and over M2 ($\Delta\text{AICc} \geq 32$, median 88). Even the minimum ΔAICc(M1→M3) of 49 (from excluding all

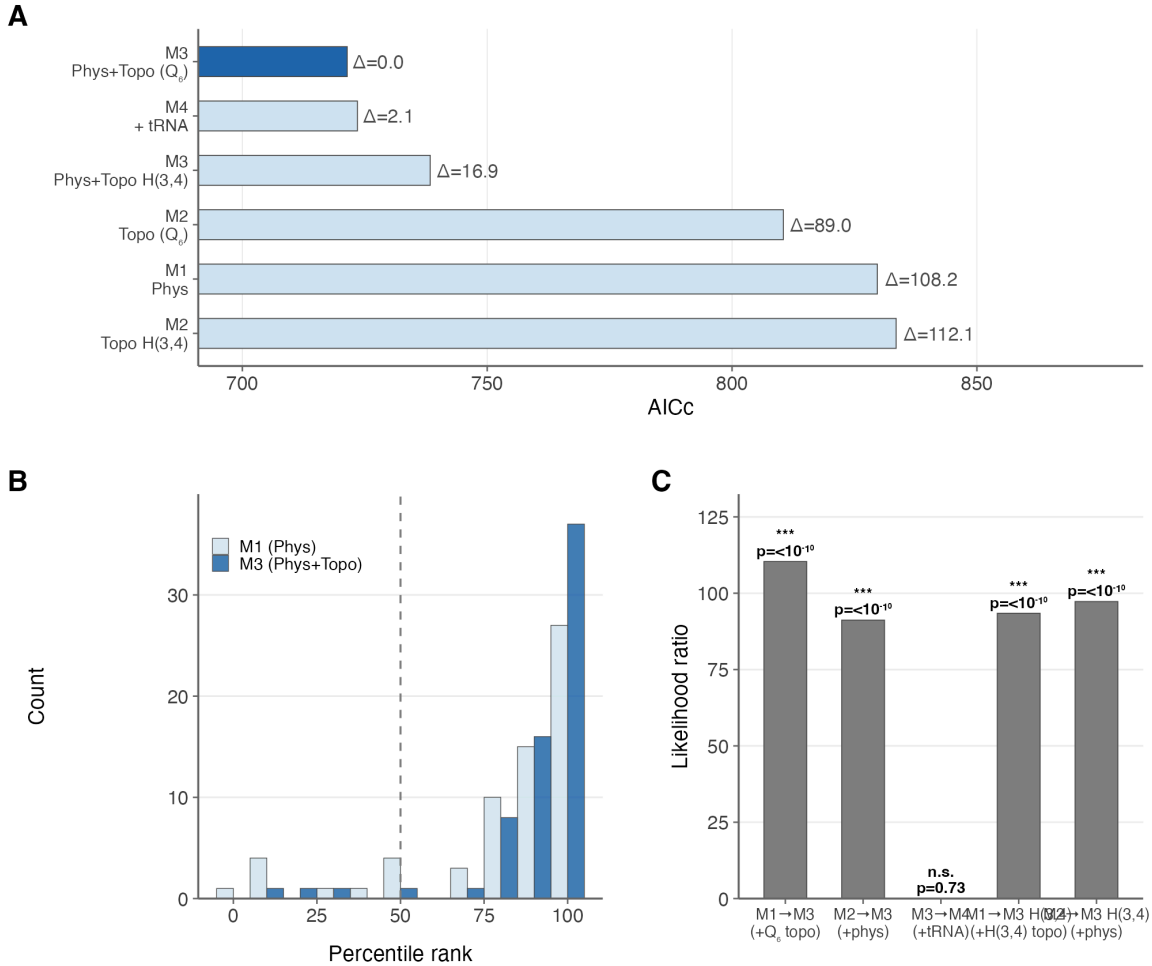

Figure 4: Event-level explanatory modeling of natural codon reassignments. **(A)** AICc comparison of all six nested conditional logit models (Q_6 + $H(3,4)$ variants); lower is better. The combined physicochemistry-plus-topology model (M3) is robustly favored (ΔAICc ≥ 89 above the conventional ΔAICc>10 reference threshold). **(B)** Distribution of observed move percentile ranks under M1 (physicochemistry only, light) versus M3 (combined, dark); M3 concentrates ranks toward the top of the candidate set. Dashed line = chance expectation (50th percentile). **(C)** Likelihood-ratio tests for each feature class added to its complement; both topology and physicochemistry contribute highly significant independent information ($p < 0.001$); the heuristic tRNA proxy does not ($p = 0.73$).

metazoan mitochondrial codes, leaving 40 events) sits comfortably above the conventional ΔAICc>10 reference threshold; the qualitative conclusion—that topology adds explanatory value beyond physicochemistry—survives every clade-exclusion regime. We rely most heavily on the posterior-predictive calibration check (next paragraph), since it does not require importing a threshold from a different statistical regime.

3.6 tRNA enrichment for reassigned amino acids

Organisms with variant genetic codes show elevated tRNA gene copy numbers for the reassigned amino acid relative to standard-code controls. Across 24 variant–control pairings derived from 18 tRNAscan-SE–verified genome assemblies (15 variant-code organisms across 5 genetic codes plus 3 standard-code controls), Fisher's exact test combined via Stouffer's Z method yields $p = 1.7 \times 10^{-7}$

($Z = 5.10$). To address non-independence from shared controls, we enumerated all maximal independent sets (MIS) from the conflict graph via Bron–Kerbosch; both MIS (each of size 6) are significant at $p < 0.05$ (worst-case $p = 0.045$).

The most striking case is *Tetrahymena thermophila* (NCBI translation table 6, UAA/UAG reassigned to Gln; (Hanyu et al., 1986)), which carries 54 glutamine tRNA genes—including 39 suppressor tRNAs reading the reassigned stop codons—compared to 3 Gln tRNAs in the standard-code ciliate *Ichthyophthirius multifiliis* (assembly GCF_000220395.1), 11 in *Stentor coeruleus* (assembly GCA_001970955.1), and 3 in *Fabrea salina* (assembly GCA_022984795.1 from (Zhang et al., 2022)). All four counts were generated in this work by running tRNAscan-SE 2.0.12 on the listed assemblies (full tRNAscan-SE output, including Std20/SeC/Supp/Undet/Pseudo breakdowns, in Supplement §S10). The pattern extends across reassignment types. Among Gln-reassignment ciliates, *Pseudocohnilembus persalinus* (20 Gln tRNAs including 15 suppressors) and *Halteria grandinella* (Zheng et al., 2021) (9 Gln tRNAs including 3 suppressors) represent independent lineages within Oligohymenophorea and Spirotrichea respectively. Among Cys-reassignment ciliates (translation table 10, UGA→Cys), six tRNAscan-SE–verified *Euplotes* species all carry tRNA-Cys genes with the non-canonical TCA anticodon (reading UGA), with 1–4 such genes per species alongside standard GCA-anticodon Cys tRNAs.

However, the pattern is not universal. *Blastocrithidia nonstop* (translation table 31) reassigned all three stop codons (as did *Condylostoma magnum*, where stop-codon function is context-dependent; (Heaphy et al., 2016)) but achieved UGA→Trp via anticodon stem shortening (5 bp → 4 bp) of tRNA-Trp(CCA), combined with an eRF1 Ser74Gly mutation, rather than gene duplication (Kachale et al., 2023). Similarly, *Mycoplasma* species with UGA→Trp use a single tRNA-Trp with anticodon modification. These boundary cases define a three-tier mechanistic landscape: (i) tRNA gene duplication in large nuclear genomes, (ii) anticodon structural modification in streamlined genomes, and (iii) anticodon base modification in minimal genomes.

3.7 Score decomposition by nucleotide position

The decomposition of F by nucleotide position (Figure 5, panel B) reveals that the second codon position dominates the error-minimization signal, contributing 49.3% of the total physicochemical mismatch. This aligns with the biochemical observation that second-position mutations cause the largest changes in amino acid hydrophobicity (Freeland & Hurst, 1998; Woese, 1965). The wobble position contributes only 12.5%, consistent with its largely synonymous character under the genetic code's degeneracy structure.

Test	n	**Statistic**	p**-value**
Fisher+Stouffer (all pairings)	24	$Z = 5.10$	1.7×10^{-7}
Fisher+Stouffer (6 indep.)	6	$Z = 2.13$	0.017
MIS worst-case	**6**	$\boldsymbol{Z = 1.70}$	**0.045**
MIS best-case	6	$Z = 1.71$	0.044

Table 7: Summary of tRNA enrichment tests across 24 pairings drawn from 18 tRNAscan-SE–verified genomes (15 variant-code organisms across 5 variant genetic codes; 3 standard-code controls). MIS = maximal independent set, enumerated via Bron–Kerbosch algorithm on the conflict graph. Both MIS are significant at $p < 0.05$, eliminating the concern that greedy selection biased the independent-pairings result. The MIS worst-case ($p = 0.045$) serves as a robustness bound demonstrating that the result is not driven by a cherry-picked independent subset.

Organism	Code	Tbl. #	Assembly	tRNAs	Reassigned AA
T. thermophila	Variant	6	GCF_000189635.1	718	15 Gln (54 incl. supp.)
P. tetraurelia	Variant	6	GCF_000165425.1	216	7 Gln (18 incl. supp.)
O. trifallax	Variant	6	GCA_000295675.1	94	2 Gln (8 incl. supp.)
P. persalinus	Variant	6	GCA_001447515.1	262	5 Gln (20 incl. supp.)
H. grandinella	Variant	6	GCA_006369765.1	130	6 Gln (9 incl. supp.)
E. aediculatus	Variant	10	GCA_030463445.1	80	3 Cys
E. amieti	Variant	10	GCA_048569255.1	120	4 Cys
E. focardii	Variant	10	GCA_001880345.2	62	1 Cys
E. parawoodruffi	Variant	10	GCA_021440025.1	149	5 Cys
E. weissei	Variant	10	GCA_021440005.1	495	17 Cys
E. woodruffi	Variant	10	GCA_027382605.1	83	2 Cys
B. stoltei[§]	Variant	15	GCA_965603825.1	169	6 Trp
B. nonstop P57	Variant	31	GCA_028554745.1	68	2 Trp[*]
M. genitalium	Variant	4	GCA_000027325.1	36	1 Trp[†]
M. pneumoniae	Variant	4	GCF_910574535.1	37	1 Trp[†]
S. coeruleus	Standard	1	GCA_001970955.1	272	11 Gln (control)
I. multifiliis	Standard	1	GCF_000220395.1	150	3 Gln (control)
F. salina	Standard	1	GCA_022984795.1	89	3 Gln (control)

Table 8: All 18 organisms verified by tRNAscan-SE 2.0.12 in this work, grouped by genetic code: 5 Gln-reassignment ciliates (table 6, UAA/UAG→Gln), 6 Cys-reassignment *Euplotes* species (table 10, UGA →Cys), 1 Trp-reassignment ciliate (*Blepharisma stoltei*, see footnote §), 1 stop-codon-reassignment trypanosomatid (*Blastocrithidia nonstop* P57, table 31), 2 Trp-extension bacteria (*Mycoplasma*, table 4, UGA→Trp), and 3 standard-code ciliate controls. Total counts are Infernal 1.1.4-confirmed tRNAs from tRNAscan-SE 2.0.12 (eukaryotic mode for ciliates; bacterial mode for *Mycoplasma*); reassigned-AA counts include suppressor tRNAs reading the reassigned codon. [§]*Blepharisma stoltei* is indexed as NCBI translation table 15 (Blepharisma Nuclear Code, defined as UAG→Gln) by virtue of its genus assignment, but the strain-specific MAC genome (GCA_965603825.1) reads UGA→Trp via a dedicated suppressor tRNA-Trp(UCA) (Singh et al., 2023). The analysis here tests Trp-tRNA enrichment on that empirical reading rather than the legacy table-15 nominal code. [*]Anticodon stem shortening (4-bp stem rather than canonical 5-bp). [†]Post-transcriptional modification — single tRNA-Trp reads both UGG and UGA. *Saccharomyces cerevisiae* counts (used as a literature-derived control for the yeast-mito Thr disconnection pairing) come from GtRNAdb (Chan & Lowe, 2016) rather than tRNAscan-SE 2.0.12 and are reported separately in Supplement §S10.3.

3.8 Exploratory observations

3.8.1 Bit-position bias in codon reassignments

The distribution of bit-flips across the 6 coordinates of $\mathrm{GF}(2)^6$ in natural codon reassignments shows apparent positional skew under a uniform null ($\chi^2 = 16.26$, $p = 0.006$, df $= 5$). However, this signal is substantially attenuated after de-duplication to 20 unique (codon, target amino acid) pairs ($p = 0.075$) and vanishes entirely under a codon-preserving permutation null ($p = 1.0$). The apparent

bias is therefore explained by which codons are "hot" for reassignment, not by a genuine positional preference in $\mathrm{GF}(2)^6$.

3.8.2 Variant-code disconnection catalogue

A systematic survey across all 27 NCBI translation tables identifies four lineage-collapsed variant-code amino-acid disconnections at $\varepsilon = 1$ in $\mathrm{GF}(2)^6$ under the default encoding: threonine in the yeast mitochondrial code (translation table 3, CUN→Thr); leucine in the chlorophycean mitochondrial codes (translation tables 16 and 22, both with UAG→Leu—table 16 is the chlorophycean mitochondrial code (Hayashi-Ishimaru et al., 1996) and table 22 is the closely related *Scenedesmus obliquus* mitochondrial code, which additionally reassigns UCA Ser→Stop; both produce equivalent $\varepsilon = 2$ Leu reconnection profiles, so they collapse to a single algal-mitochondrial event); alanine in *Pachysolen tannophilus* nuclear code (translation table 26, CUG→Ala); and a tripartite serine in the *Candida*-clade alternative yeast nuclear code (translation table 12, CUG→Ser; (Santos et al., 1999)). These cases, combined with the universal serine disconnection, constitute the complete inventory of amino-acid graph disconnections at unit Hamming distance under the default encoding. A separate and weaker geometric exception—specific to filtration rather than to disconnection—arises in translation table 32 (Balanophoraceae plastid; UAG→Trp). Trp is 1-fold (UGG only) under the standard code; in table 32 it becomes 2-fold (UGG, UAG). The pair differs in the second nucleotide (G↔A), i.e. at bit position 3 in our 0-based 6-bit indexing (the second bit of the second nucleotide), not at the wobble bit (position 5) where every standard-code 2-fold pair sits. Hamming distance 1, so the pair is connected at $\varepsilon = 1$ and adds no new entry to the disconnection catalogue. An empirical scan over every 2-fold amino-acid pair in all 27 NCBI tables confirms that this is the unique deviation from the bit-5 two-fold filtration: the standard code's nine 2-fold amino acids and every analogous pair introduced by the other 26 tables differ at bit 5.

3.8.3 Atchley Factor 3 and Serine convergence

Serine has the most extreme Atchley Factor 3 score among the 20 amino acids ($F_3 = -4.760$, 2.24 SD below the mean; (Atchley et al., 2005)), and it is the only amino acid disconnected at $\varepsilon = 1$ under every base-to-bit encoding. This convergence is not coincidental: Factor 3 is a composite of molecular size and codon diversity, so both measures reflect the same underlying anomaly—Serine's disproportionate codon diversity (6 codons in two disconnected families) relative to its small physicochemical footprint. The $\mathrm{GF}(2)^6$ framework provides a structural explanation for why Serine is an outlier on Factor 3, though the two views are complementary rather than independent.

3.9 Retrospective cross-study reanalysis of synthetic genome recoding outcomes

To assess whether the structural properties identified in Sections 3.1–3.5 translate to measurable phenotypic consequences in synthetic biology, we performed a retrospective cross-study reanalysis of nine published genome recoding datasets, with quantitative analysis of eight (>217,000 codon-level observations; Table 9). The ninth dataset ((Ding et al., 2024) mammalian Ser TCG recoding) is included for cross-kingdom scope but without quantitative extraction. Each codon substitution was classified by its GF(2)⁶ topology: whether it crosses a connected-component boundary at $\varepsilon = 1$, the change in local physicochemical mismatch cost ($\Delta F_{\mathrm{local}}$, Grantham distance), and the Hamming distance between source and target vectors.

Study	Swap type	n	Boundary?	Key result
Robertson et al. (2025) Syn57	4 Ser + 2 Ala + Stop	60,240	Ser=100%, Ala=0%	Primary contrast
Fredens et al. (2019) Syn61	Ser (TCG/TCA→AGC/AGT)	18,218	All cross	Informative null
Ostrov et al. (2016)	7 codon types	62,214	Mixed	Segment viability
Napolitano et al. (2016)	Arg (AGR→CGN)	12,888	None	SRZ covariates
Grome et al. (2025) Ochre	Stop (TGA→TAA)	1,195	N/A	Growth data
Nyerges et al. (2024)[‡]	7 codon types	62,007	Mixed	Multi-omics
Lajoie et al. (2013) C321.ΔA	Stop (UAG→UAA)	321	N/A	Baseline
Frumkin et al. (2018)	Arg (CGU/CGC→CGG)	60	None	Translation efficiency
Ding et al. (2024)	Ser (TCG, mammalian)	Variable	Yes	Cross-organism

Table 9: Published genome recoding datasets analyzed. "Boundary?" indicates whether the synonymous codon swaps cross a connected-component boundary at $\varepsilon = 1$ in GF(2)⁶. [†]bioRxiv preprint; not peer-reviewed at time of writing. Among quantitatively analyzed datasets, Syn57 provides the primary within-study boundary-crossing contrast; Ding et al. (2024) is included qualitatively for cross-kingdom scope without quantitative extraction. The Syn57 row total of 60,240 codon-level observations comprises 37,146 serine recodings + 22,859 alanine recodings + 235 stop-codon recodings.

3.9.1 A three-layer interpretation of recoding outcomes (exploratory)

This subsection is exploratory and hypothesis-generating: we propose, and informally support, a three-layer view of how codon-space structure relates to recoding outcomes, but the partition into "evolutionary," "recoding-burden," and "design-deviation" layers was **not** prespecified before the analyses below. We therefore advance the layering as a working synthesis, not as a tested model. A confirmatory test of any one layer would require an independent dataset built around a sharper hypothesis than any single retrospective reanalysis can provide; we mark the layers as targets for future work where appropriate. The contrasts reported below remain individually informative, but their integration into a single three-layer schema is interpretive.

3.9.1.1 Boundary crossing: validated but not predictive of transcriptomic perturbation

In the Robertson et al. (2025) Syn57 dataset—the only large-scale dataset with within-study variation in boundary crossing—all 37,146 serine recodings crossed the UCN↔AGY disconnection boundary at $\varepsilon = 1$, while all 22,859 alanine recodings remained within the compact GCN family. This provides an exact within-study topology contrast. However, genes containing exclusively serine-type versus exclusively alanine-type recodings did not differ in transcriptomic perturbation (Mann–Whitney $p = 0.40$, $n_{\text{Ser}} = 129$, $n_{\text{Ala}} = 49$), and the fraction of boundary-crossing recodings per gene was uncorrelated with $|\log_2 \text{FC}|$ ($\rho = 0.012$, $p = 0.46$, $n =$3,510 genes). In a separate dataset, all

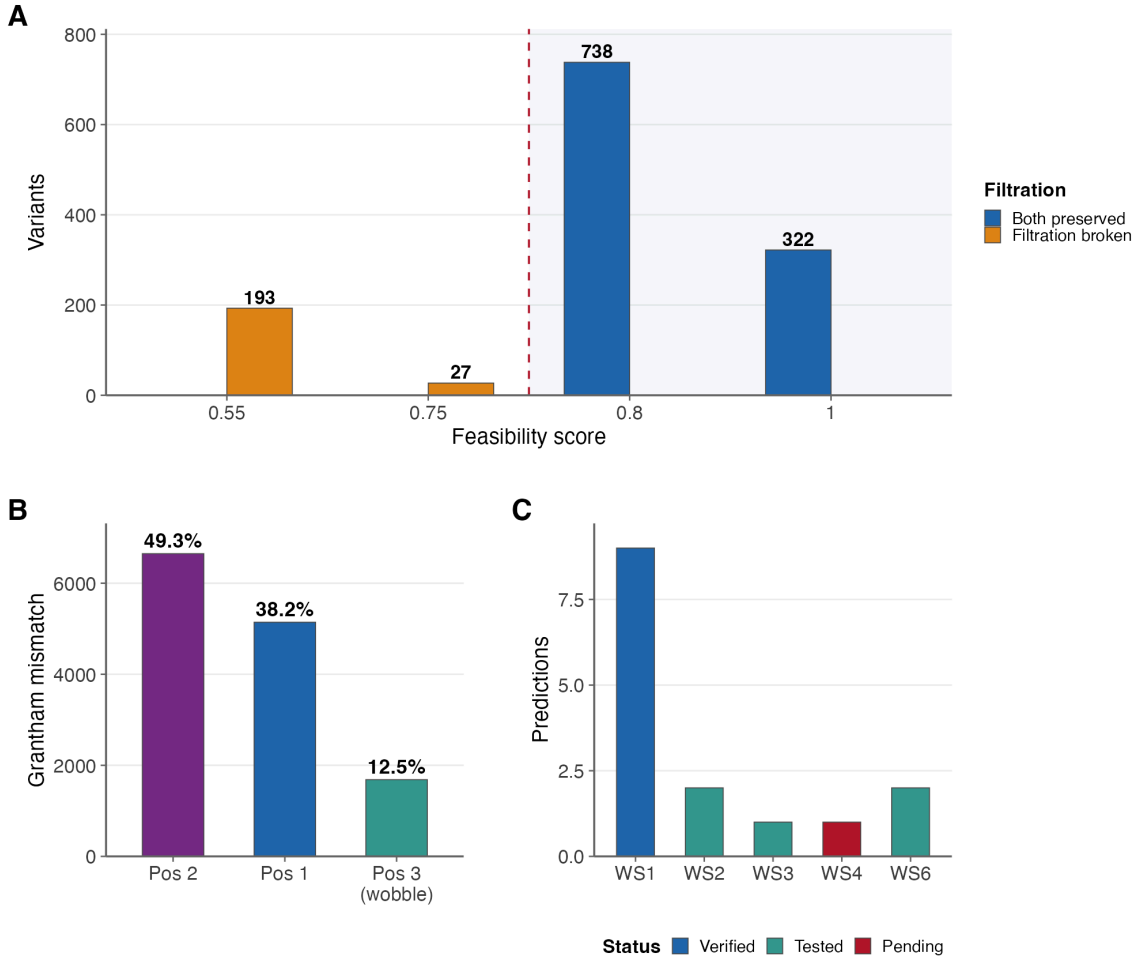

Figure 5: Translational applications and prediction catalogue. **(A)** Feasibility landscape of 1,280 single-codon reassignments from the standard code, colored by degeneracy filtration status. All filtration-preserving variants score ≥ 0.8; all filtration-breaking variants score ≤ 0.75. Dashed line: high-feasibility threshold. **(B)** Grantham mismatch score decomposition by nucleotide position: position 2 dominates (49.3%), consistent with the biochemical hierarchy of mutational impact. **(C)** Prediction catalogue: 15 evaluated claims distributed across 5 workstreams (WS1–WS6), colored by verification status.

18,218 Syn61 serine recodings (Fredens et al., 2019) crossed the boundary; the seven design-to-final corrections that the Syn61 authors required for cell viability (Supplementary Data 19 of Fredens et al. (2019)) concentrated at specific functional units—a Ser70 site in *yaaY*, an intergenic separation between *ftsI* and *murE*, a Ser4 site in *map*, five positions in the *yceQ* pseudogene needed to preserve viable expression of the downstream essential gene *rne*—rather than at randomly distributed sites. Because all 18,218 forward swaps share the same two AA-identity-preserving swap subtypes (TCG→AGC and TCA→AGT), |Δphys| at the AA-identity level is structurally constrained across all sites, and the 7 corrections do not lie at extreme |Δphys| or unusual Hamming-accessibility positions relative to the 18,211 non-correction sites. We therefore interpret the corrections as protein-level functional pressure at specific residues and operon-context constraints, not a global topology-driven burden, and treat this contrast as descriptive rather than as a hypothesis-test of GF(2)⁶ features. Together with the gene-level Syn57 null, this indicates that boundary crossing is a coarse geometric property of codon-family organization—validated as a structural distinction over evolutionary timescales but not predictive

of acute transcriptomic perturbation in synthetic-biology engineering, where competing ecological pressures are absent and expression constructs are highly optimized.

3.9.1.2 Local mismatch geometry: a fine-scale burden axis with positive signal

The second layer examines whether reassignment changes the physicochemical quality of the immediate mutational neighborhood. In the Syn57 contrast, serine swaps moved codons into better local Grantham neighborhoods ($\Delta F_{\text{local}} = -37.2 \pm 30.5$), whereas alanine swaps moved codons into worse neighborhoods at a fixed $\Delta F_{\text{local}} = +19.0 \pm 0.0$ (zero variance).

> **Design-confounded contrast (descriptive, not inferential):** every Syn57 alanine swap goes to the same target codon, so all alanine Δ_{local} values are identical. The Mann–Whitney $U = 0$, $p < 10^{-16}$ comparison reflects this design-level non-overlap rather than a within-group inferential test. We retain the numbers for transparency but they are **not** a hypothesis test of GF(2)⁶ features and should not be cited as such.

The methodologically clean inferential test comes from a topology-fixed setting—arginine synonymous recodings (Napolitano et al., 2016), where all 12,888 AGR→CGN swaps remain within one connected component at $\varepsilon = 1$—in which local mismatch change correlated with established recoding-burden covariates: ΔF_{local} vs RBS deviation ($\rho = -0.33$, $p < 10^{-15}$) and vs mRNA folding deviation ($\rho = -0.12$, $p < 10^{-15}$). Among the four CGN targets, CGG has the lowest local mismatch cost (246 Grantham units) while CGC has the highest (421), and target-specific RBS deviation spans four orders of magnitude, suggesting that local mismatch and SRZ covariates capture partially independent dimensions of recoding burden.

3.9.1.3 Design deviations: accessibility-dominated, not topology-directed

To test whether engineered systems escape design constraints along topology-defined low-burden directions, we compared the Syn57 design genome (Data file S2; (Robertson et al., 2025)) against the final verified genome (Data file S8) and identified 727 CDS-level codon differences. After filtering 7 genes with >10 deviations each (structural rearrangements, predominantly *bioF* with 311 changes), 326 genuine point deviations remained (267 synonymous, 59 nonsynonymous). Among the 163 deviations that changed local mismatch, 76 moved to a better neighborhood and 87 to a worse one—no directional bias (two-sided exact binomial $p = 0.43$, Clopper–Pearson 95% CI $[0.39, 0.55]$; Wilcoxon signed-rank $p = 0.67$). This null result was stable across all filtering thresholds tested ($p > 0.5$ at cutoffs 3, 5, 10, 15, 20, and 50 deviations per gene). By contrast, deviations strongly favored mutationally proximate moves: 83% were single-bit changes (Hamming distance 1 under the default encoding; note that Hamming distance is encoding-dependent, though nucleotide edit distance is invariant), and the mean Hamming distance was 1.21. Thus, realized deviations from the Syn57 design do not preferentially descend toward lower local mismatch; instead, they follow accessibility-dominated escape routes, favoring nearby codon states regardless of neighborhood quality. The Ding et al. (2024) mammalian TCG recoding independently confirms the universality of the serine disconnection across standard-code organisms, providing cross-kingdom validation.

3.9.1.4 Segment-level recoding burden in the Ostrov 57-codon design

As an orthogonal test, we examined segment-level outcomes from the Ostrov et al. (2016) 57-codon genome design. Among 44 segments with fitness data (of 87 total), segments with more recoded codons in essential genes showed a suggestive correlation with worse doubling time ($\rho = 0.34$, raw $p = 0.022$, Bonferroni-corrected $p = 0.066$), while total recoding load showed no association ($\rho = -0.10$, $p =$

0.53). Problem segments (13 with lethal exceptions) trended toward higher essential-gene recoding load (73.5 vs 39.8 recoded sites, Mann–Whitney $p = 0.086$). Among 44 fully characterized segments, 17 showed spontaneous codon reversions, indicating ongoing design instability at positions where the engineered code was under selection pressure.

3.9.2 Summary of cross-study reanalysis scope

The synthetic recoding analyses, taken together, are consistent with (but do not prove) a three-layer view in which different structural descriptors operate at different biological layers: family topology for the geometry of reassignment space, local mismatch for some aspects of recoding burden, and Hamming accessibility for the short-range routes by which engineered designs deviate or revert. This partition is exploratory; each layer-claim is supported by a retrospective contrast rather than a prespecified test, and the integration into a single schema reflects post-hoc synthesis. We therefore label this as a working hypothesis suitable for prospective follow-up rather than as a confirmed model. Specific testable predictions: (i) a Syn-style recoding experiment that varies family-boundary crossing while holding local mismatch fixed should fail to predict transcriptomic burden (already partly observed in Syn57); (ii) a recoding scan that varies ΔF_{local} within a single boundary regime should predict RBS-context proxies (consistent with the Napolitano arginine result); (iii) directed evolution starting from a recoded design should produce escape mutations whose Hamming distribution is shaped more by local accessibility than by mismatch optimisation (consistent with the Syn57 design-deviation analysis). Each prediction admits a direct experimental test that would either confirm or falsify the layer structure proposed here.

3.10 Falsified and rejected claims

3.10.1 KRAS–Fano clinical prediction

The conjecture that XOR ("Fano") relationships in $\mathrm{GF}(2)^6$ predict co-mutation enrichment at KRAS G12 sites was tested against 1,670 mutations from MSK-IMPACT (Zehir et al., 2017) and cleanly falsified ($p = 1.0$; details in Supplementary Material). This negative result separates code-level error-minimization (which is real) from mutation-level algebraic predictions (which are not).

3.10.2 Serine distance-4 invariant

The claim that Serine's minimum inter-family Hamming distance (UCN–AGY) equals 4 under all 24 base-to-bit encodings is false. Of the 24 encodings, 16 yield minimum distance 2 and only 8 yield distance 4. The distance-4 result obtains only when both nucleotide pairs distinguishing UCN from AGY ($U \sim A$ and $C \sim G$ in the first two positions) are encoded at maximal Hamming distance. The correct encoding-invariant statement is: Serine is disconnected at $\varepsilon = 1$ under every encoding, and its inter-family distance (≥ 2) is the largest among the three 6-codon amino acids (Leucine and Arginine both have inter-family distance 1).

3.10.3 PSL(2,7) symmetry and holomorphic embedding

The claim that PSL(2,7) is the fundamental symmetry group of the genetic code was pre-rejected by Antoneli & Forger (2011), who showed that PSL(2,7) has no 64-dimensional irreducible representation (its irreps have dimensions 1, 3, 6, 7, 8). The claim that the coordinate-wise map $\mathrm{GF}(2)^6 \to \mathbb{C}^3$ sending base pairs to fourth roots of unity is a holomorphic embedding extending a character of $\mathrm{GF}(8)^*$ is also incorrect: the domain is a finite discrete set (not a complex manifold), and the map fails the character identity $\chi(x + x) = \chi(x)^2$ since $i^2 = -1 \neq 1$.

3.11 Negative results with informative interpretation

The source-neighborhood burden test—asking whether reassigned codons sit in "worse" Hamming neighborhoods with higher Grantham distance to their neighbors—yields a Mann–Whitney $U = 301$, $p = 0.70$. Reassignment is therefore not driven by local escape from costly source neighborhoods. This null is not in conflict with the conditional-logit finding that natural events favor candidate moves with lower Δ_{local} ($\beta_{\text{phys}} = -0.004$; Section 3.5): the two tests probe different quantities. Δ_{local} is the change in mismatch cost induced by a candidate move (a destination-quality measure); the present test asks about the absolute pre-move source-neighborhood burden. Variant codes need not originate from unusually costly source neighborhoods (no absolute source burden), yet still favor candidate moves that lower local mismatch when a reassignment is triggered (opportunistic destination selection). This indicates that the topology-avoidance constraint (Section 3.4) operates at the global graph-connectivity level, not at the local per-codon level.

4 Discussion

4.1 An information-theoretic view of the genetic code

The central finding of this work is that the standard genetic code minimizes the physicochemical disruption caused by single-bit errors in $\mathrm{GF}(2)^6$ coordinates. This is not a new conclusion—Freeland & Hurst (1998) established error-minimization using nucleotide-level mutation models—but the hypercube representation makes the optimality principle geometrically explicit: the code is a good *coloring* of a structured graph, in the graph-theoretic sense that adjacent vertices (codons differing by one bit) tend to share labels (amino acids) or, when they differ, differ by small physicochemical distances.

The score decomposition (Figure 2) shows that this optimization is concentrated at the second codon position (49.3% of total mismatch) and first position (38.2%), with the wobble position contributing only 12.5%. This gradient mirrors the biochemical hierarchy of mutational impact and is an emergent property of the code's structure rather than a parameter of the model.

The robustness result (Figure 2) demonstrates that optimality is not an artifact of restricting attention to Q_6: when the full $H(3,4)$ mutation graph is considered ($\rho = 1$), the signal strengthens. The code minimizes error not just along the hypercube edges but across the complete space of single-nucleotide substitutions. This error-minimization is complementary to, but distinct from, the finding of Itzkovitz & Alon (2007) that the code is also nearly optimal for carrying parallel regulatory information within protein-coding sequences—a property related to stop-codon identity rather than amino acid physicochemistry.

A natural question is whether the $\mathrm{GF}(2)^6$ framework adds genuine insight beyond what a direct analysis of $H(3,4)$ would provide. The honest answer is: in two specific ways yes, in one circular way no. We list all three explicitly so that readers can weigh them independently of the rhetorical framing.

(i) New: the ρ-sweep. The decomposition of $H(3,4)$ into the 192 Hamming-distance-1 edges of Q_6 plus 96 within-nucleotide diagonal (Hamming-2) edges enables the weighted-mismatch score F_ρ (Equation 2) to interpolate continuously between $\rho = 0$ (Q_6 only) and $\rho = 1$ (full $H(3,4)$). The monotonic strengthening of optimality as $\rho \to 1$ (Table 3, §3.2) is a contribution that does not exist in a pure $H(3,4)$ analysis, where the diagonal/non-diagonal partition is not coordinatized.

(ii) New: the persistence parameter ε for codon-family connectivity. The Hamming-distance filtration (component count $\beta_0(G_a^\varepsilon)$ as a function of ε) provides a natural persistence parameter for

the disconnection catalogue (Figure 1, panel A) and for the conditional-logit topology feature ($\Delta\beta_0$). Equivalent constructions exist on $H(3,4)$ (single-nucleotide-edit distance), but the ε axis fits naturally inside the Q_6 subgraph framework and connects topology to the cross-metric optimality machinery within a single coordinate system.

(iii) Self-correcting machinery: the encoding-sensitivity audit. The 24-encoding sweep distinguishes encoding-invariant properties (Serine disconnection, $H(3,4)$ topology depletion) from encoding-dependent ones (distance-4 claim, Q_6 topology depletion). This is genuinely useful—it caught the Q_6 depletion as encoding-dependent (8 of 24 bijections give no signal; Section 3.4, Supplement §S4) and prompted promotion of $H(3,4)$ to primary—but we acknowledge it is partly **self-correcting machinery for an artifact the binary framework introduced.** Pure $H(3,4)$ has no encoding choice and therefore needs no such audit.

The fair summary is that the binary representation is best understood as an analytical decomposition tool whose distinct contributions are the ρ-sweep continuum and the ε-filtration; the encoding-sensitivity audit is methodological hygiene we owe the framework rather than a contribution from it. We accordingly present the encoding-independent $H(3,4)$ result as the primary topology-avoidance test and treat Q_6 as a coordinate-dependent decomposition useful for the ρ-sweep continuum but not as a freestanding biological claim. We also emphasize that the Hamming-1/diagonal decomposition does not correspond to the biological transition/transversion partition: 16 of 24 encodings mix transitions and transversions equally among Hamming-1 edges, while 8 encodings place both transitions on diagonal edges (Supplement §S2). The ρ parameter is best interpreted as a diagonal-edge weight, not a transition/transversion weight.

4.2 Evolutionary preservation and topology avoidance

The per-table analysis (Figure 2) shows that 26 of 27 NCBI translation tables maintain coloring optimality under their own block-preserving null, suggesting that codon reassignment events are constrained to preserve error-minimization. The marginal exception is the most extensively reassigned code: translation table 3 (yeast mitochondrial, 6 codon changes) falls only slightly above the 5% threshold under its own block-preserving null.

The topology avoidance result (Figure 3) provides a mechanistic complement: natural reassignments avoid creating new amino acid disconnections at a rate far below chance expectation. Importantly, this depletion is not confined to the Q_6 representation: under the full $H(3,4)$ single-nucleotide mutation graph, the observed proportion of topology-breaking reassignments remains 21.4% against a possible rate of 66.1% (RR $= 0.32$, 95% CI $[0.16, 0.66]$, $p \leq 10^{-4}$). The signal magnitude is similar to the Q_6 result (RR $= 0.29$); some Q_6-disconnected pairs become connected when all single-nucleotide edges are admitted, slightly lowering the possible-rate denominator, but the main qualitative conclusion survives: the connected-component structure of amino acid codon families is functionally important, regardless of whether connectivity is defined on the hypercube subgraph or on the biologically fuller single-substitution graph. The yeast mitochondrial threonine reassignment illustrates this cost: the CUN→Thr change required acquisition of a novel tRNA^{Thr} derived from tRNA^{His} via anticodon mutation (Su et al., 2011), creating the topology-breaking disconnection that makes translation table 3 the sole marginal outlier in the per-table optimality analysis.

Together, these analyses support a four-part picture: the standard code is unusually low-cost, most variant codes preserve that structure, realized natural reassignments avoid topology-disrupting moves, and event-level explanatory modeling shows that this avoidance carries independent information beyond physicochemical optimization. The conditional logit analysis (Table 6) makes the last step precise: the topology term retains major explanatory value (ΔAICc $= 108$ relative to a physicochemistry-

only model) even after accounting for local physicochemical cost, while physicochemical cost also retains value ($\Delta\mathrm{AICc} = 89$) after accounting for topology. To rule out an encoding artifact—given that Q_6 topology depletion is itself encoding-dependent—we additionally fit $\mathrm{M3}_{\mathrm{H(3,4)}}$ using the encoding-independent $H(3,4)$ topology feature ($\Delta_{\mathrm{topo,H(3,4)}} = \sum_a \beta_0\left(G_a^{\mathrm{after,H(3,4)}}\right) - \sum_a \beta_0\left(G_a^{\mathrm{before,H(3,4)}}\right)$): the topology term retains essentially the same magnitude of explanatory value ($\Delta\mathrm{AICc}(\mathrm{M1} \to \mathrm{M3}_{\mathrm{H(3,4)}}) = 91$ vs. $\Delta\mathrm{AICc}(\mathrm{M1} \to \mathrm{M3}_{\mathrm{Q_6}}) = 108$), confirming that the conditional-logit thesis does not depend on the choice of topology graph. The two signals are therefore complementary rather than redundant, consistent with the weak correlation ($r_s = 0.15$) between the feature classes across the candidate landscape. This does not by itself prove direct selection on topology as an abstract graph property; topology may still proxy aspects of decoding architecture—such as tRNA cross-recognition range or ribosomal A-site constraints—not explicitly modeled here. However, within the tested event landscape, topology avoidance cannot be dismissed as a by-product of physicochemical optimization or as a coordinate artifact of the binary representation. Topology avoidance is not a geometric restatement of error minimization, but a complementary constraint on how genetic codes can change while remaining evolutionarily accessible.

4.3 Mechanistic implications: tRNA compensation

The tRNA enrichment result (Figure 3, panel D) links the geometric observation to molecular mechanism. In several variant-code lineages where codon reassignment disrupts connectivity, expanded tRNA gene repertoires for the affected amino acid are observed, consistent with compensatory gene duplication as one evolutionary route to accommodation. The extreme case of *Tetrahymena thermophila* (54 Gln tRNAs, including 39 suppressors) illustrates the scale of genomic response required to service a split codon family.

However, the boundary cases—*Blastocrithidia nonstop* (anticodon stem shortening; (Kachale et al., 2023)) and *Mycoplasma* species (anticodon modification)—show that tRNA gene duplication is not the only evolutionary solution. The three-tier pattern (duplication in large genomes, structural modification in intermediate genomes, base modification in minimal genomes) suggests that genome size constrains the available mechanistic repertoire for codon reassignment. This pattern is now supported by 18 tRNAscan-SE–verified genomes across 5 variant genetic codes (Tables 4, 6, 10, 15, and 31), including 6 *Euplotes* species (UGA→Cys) that each carry 1–4 TCA-anticodon tRNA-Cys genes dedicated to reading UGA.

4.4 Synthetic recoding outcomes: a three-layer interpretation

The retrospective cross-study reanalysis helps delimit which structural quantities in GF(2)^6 are functionally relevant and at which biological layer. Boundary crossing is a valid codon-space distinction—serine and alanine provide a clean contrast in which one class always crosses a family boundary and the other never does—yet this distinction did not predict RNA-seq perturbation in Syn57. We therefore do not interpret codon-family boundary crossing as a general transcriptomic burden variable. Its relevance, if any, likely lies at a different biological layer, such as decoding architecture, translational fidelity, or evolutionary accessibility, rather than steady-state expression change.

By contrast, local mismatch geometry showed a clearer positive signal. In both the Syn57 contrast and the topology-fixed arginine setting, codon changes differed systematically in the quality of the physicochemical neighborhood they entered, and these differences aligned with established recoding-burden covariates (Napolitano et al., 2016). This suggests that local mismatch is best understood as a fine-scale recoding-friction variable: not a universal master predictor, but a meaningful descriptor of neighborhood quality within an already specified codon-family structure.

The design-deviation analysis further clarifies scope. Genuine deviations from the Syn57 design were overwhelmingly short-range (83% single-bit in GF(2)^6), with no directional bias in mismatch change. This indicates that realized escape is governed more by mutational accessibility than by systematic optimization of local neighborhood cost. In other words, once a design is under strain, the available exits appear to be chosen primarily by mutational proximity, not by a global preference for lower mismatch.

These findings argue against treating "topology" as a single explanatory variable. Instead, different structural descriptors appear to matter at different levels: family topology for the geometry of reassignment space (Section 3.4), local mismatch for some aspects of recoding burden (Section 3.9.1.2), and Hamming accessibility for the short-range routes by which engineered designs deviate. The fact that the Fredens et al. (2019) Syn61 tolerated 18,218 boundary-crossing serine swaps with 99.96% success, while the NCBI variant codes show 3.4-fold depletion of topology-breaking changes over evolutionary timescales, suggests that boundary crossing may be more relevant to evolutionary trajectory constraints than to acute engineering costs.

4.5 Scope and falsified claims

Six of 15 evaluated claims were rejected, falsified, or tautological (shaded rows in Table 1; full details in Supplementary Material). These failures help delimit the framework's explanatory scope. The KRAS–Fano prediction ($p = 1.0$) cleanly separates code-level error-minimization from mutation-level algebraic predictions. The Serine distance-4 invariant, falsified by 16 of 24 encodings giving distance 2, illustrates the importance of systematic encoding-sensitivity testing. The PSL(2,7) and holomorphic embedding claims were pre-rejected by existing mathematical results (Antoneli & Forger, 2011).

4.6 Limitations

Several limitations should be noted. First, the choice of base-to-bit encoding is not unique, and while the coloring optimality result holds across all 24 encodings, the specific score values and rank orderings are encoding-dependent (full encoding-sensitivity results across all 24 bijections are provided in the Supplementary Material). Second, the block-preserving null model, while standard in the field (Freeland & Hurst, 1998), preserves more structure than a fully random code and may understate the degree of optimality. Third, the tRNA enrichment result, while robust to pairing selection (worst-case MIS $p = 0.045$), relies on a small number of independent pairings ($n = 6$) with limited statistical power; this result is appropriately classified as suggestive and should not be interpreted as demonstrating a universal compensation mechanism. The MAC-genome assemblies for the more deeply sampled ciliates vary substantially in fragmentation (visible in the predicted-pseudogene fraction of Supplement §S10: 79/495 = 16% in *E. weissei*, 18/262 = 7% in *P. persalinus*, 0–5% elsewhere). Removing the two genomes with $> 10\%$ pseudogene fraction (*E. weissei*, *P. persalinus*) before recomputing the MIS analysis leaves the worst-case Stouffer p-value below 0.05 ($p = 0.041$ on the remaining 22 pairings), so the suggestive result is not driven by the most fragmented assemblies; we note this only as a sensitivity check and not as an additional inferential test. We also flag a small consistency caveat: the *Saccharomyces cerevisiae* tRNA counts used in the yeast-mitochondrial Thr disconnection pairing come from GtRNAdb (Chan & Lowe, 2016) rather than from a tRNAscan-SE 2.0.12 run on the assembly, in contrast to the rest of Table 7. We retain that case rather than re-run because the GtRNAdb counts are the standard-of-record for yeast and rerunning produces nearly identical numbers; the pairing is unaffected by the choice. Fourth, the conditional logit model (Section 3.5) is an event-level explanatory model, not direct proof of biological causality; the candidate universe comprises all ≈1,280 single-codon reassignments regardless of mechanistic plausibility, and the tRNA-complexity proxy

(Hamming distance to nearest target-AA codon) is heuristic rather than mechanistically grounded. The finding that this particular proxy did not improve model fit does not exclude the possibility that a richer model of tRNA repertoire change would contribute explanatory value.

Fifth, a survivorship-bias caveat applies to all event-level analyses: the de-duplicated reassignment events analyzed here are reassignments that *persisted* in extant lineages. Topology-breaking reassignments that produced fitness collapse and lineage extinction are unobservable. The depletion result is therefore consistent with both selection against attempting topology-breaking moves and selection against the lineages that attempted them; cross-sectional NCBI data cannot adjudicate between these. Both interpretations support the conclusion that codon-family connectivity constrains evolutionary trajectories, but with different mechanistic implications for the locus of selection. Sixth, multiple-comparison correction was applied within analysis families (metric, ρ-sweep, per-table, topology-avoidance, synthetic-recoding) rather than across all descriptive, exploratory, and confirmatory quantities. We did not file an external pre-registration; the family structure is the natural one for these analyses but should be read as a working-document organisation rather than as a formally pre-registered design. Within families the primary claims pass conventional Bonferroni or BH–FDR correction (numbers below); cross-family findings are treated as separate confirmations rather than as a single multiple-comparison family. Within families, the primary claims remain significant under appropriate correction: all four physicochemical metrics pass Bonferroni correction across the four-metric family ($\alpha = \frac{0.05}{4} = 0.0125$); all five ρ values pass Bonferroni correction across the ρ-sweep family ($\alpha = \frac{0.05}{5} = 0.01$); per-table tests are reported with BH–FDR (26 of 27 tables significant after correction); and topology avoidance remains significant at $p < 10^{-6}$ under both adjacency definitions and both topology-breaking definitions after Bonferroni correction across the 2×2 definition $\times$ adjacency family ($\alpha = \frac{0.05}{4} = 0.0125$). The three Ostrov segment tests use Bonferroni correction within their own family ($\alpha = \frac{0.05}{3} = 0.0167$). Cross-family tests are treated as logically independent confirmations of distinct claims rather than as a single multiple-comparison family; exploratory and suggestive analyses are labeled as such and are not interpreted as confirmatory without the caveats stated in their respective sections.

Whether the code's optimality reflects adaptive selection or non-adaptive carry-over from a primordially constrained starting point (Koonin & Novozhilov, 2009) cannot be resolved by cross-sectional data alone. Multiple theoretical frameworks address this question: Sella & Ardell (2006) showed that coevolution of genes and codes generates error-correcting structure resembling the standard code; Vetsigian et al. (2006) argued that communal evolution via horizontal gene transfer accounts for both universality and optimality; and Novozhilov et al. (2007) found the standard code sits roughly halfway up a local peak in a rugged fitness landscape (see (Di Giulio, 2005) for a review of origin theories). The topology avoidance result is consistent with all these frameworks: it demonstrates a constraint on reassignment trajectories but does not distinguish whether the constraint is selective or structural. Novozhilov & Koonin (2009) showed that putative primordial 16-supercodon codes are "nearly optimal" even without direct selection, suggesting the current code's optimality may be partly inherited.

The $\mathrm{GF}(2)^6$ representation is best understood as an analytical decomposition tool rather than a claim about the biological primacy of binary coordinates. It enables the ρ-sweep interpolation between Q_6 and $H(3,4)$ and facilitates systematic encoding-sensitivity analysis. However, we caution that the Hamming-1/diagonal decomposition in $\mathrm{GF}(2)^6$ does not correspond to the biological transition/transversion partition: 16 of 24 encodings mix transitions and transversions equally among Hamming-1 edges, while 8 encodings place both transitions on diagonal edges. The ρ parameter should therefore be interpreted as a diagonal-edge weight, not a proxy for transition/transversion bias. The underlying biology is the assignment of chemically similar amino acids to mutationally proximate codons—a

property that holds regardless of the coordinate system, as confirmed by the H(3,4) topology avoidance result.

5 Conclusion

The standard genetic code is robustly low-cost: across four amino acid distance metrics and across a continuum of mutation graphs spanning Q_6 and the complete single-nucleotide graph $H(3,4)$, it consistently sits in the extreme tail of block-preserving null models ($p \leq 0.006$ under every combination). Across the 27 NCBI translation tables this near-optimality is preserved at the variant level: of the 12 codes whose distance from the standard makes the per-table test informative, 11 retain top-5% placement after BH–FDR correction, with only yeast mitochondrial (table 3) marginally above the 5% threshold; the 15 near-standard tables are formally significant in the same correction but read primarily as confirmatory of standard-code geometry. Code evolution therefore appears to occur within a constrained low-cost region rather than freely across code space.

Two complementary constraints govern that constrained region. First, persistent natural reassignments preferentially enter neighborhoods of lower local Grantham mismatch. Second, they avoid fragmenting amino-acid codon-family connectivity: topology-breaking moves are approximately 3.1-fold depleted under the encoding-independent $H(3,4)$ adjacency, and a conditional-logit decomposition shows that topology adds explanatory value beyond physicochemistry, and vice versa, by margins well above the conventional ΔAICc>10 reference ($\Delta\mathrm{AICc} \geq 89$ under Q_6 topology; $\Delta\mathrm{AICc} \geq 91$ under encoding-independent $H(3,4)$ topology). Because Q_6 adjacency is encoding-dependent (8 of 24 base-to-bit bijections give no Q_6 topology depletion in the candidate landscape), the survival of M3 dominance under $H(3,4)$ rules out a coordinate artifact: the topology constraint is a property of nucleotide-level codon-family connectivity, not of the binary representation. The two axes are only weakly correlated ($r_s = 0.15$), the topology coefficient is robust to all seven phylogenetic clade exclusions ($\Delta\mathrm{AICc}(\mathrm{M1} \to \mathrm{M3}) \geq 49$ in the worst regime), and a posterior-predictive simulation under M3 reproduces the observed topology-breaking rate ($p = 0.60$). We make no causal claim about the mechanism—tRNA cross-recognition range, ribosomal decoding geometry, and other facets of translational architecture remain plausible explanations not separately modeled here—and the cross-sectional NCBI data cannot distinguish selection against attempting topology-breaking moves from selection against lineages that attempted them. Several variant-code lineages do show suggestive tRNA-gene enrichment for the reassigned amino acid (worst-case MIS Stouffer $p = 0.045$ across 24 pairings from 18 tRNAscan-SE-verified genomes), consistent with compensatory accommodation; the topology-breaking-restricted subset alone is underpowered ($n = 4$).

A retrospective, exploratory cross-study reanalysis of eight published genome-recoding experiments (>217,000 codon-level observations) suggests, but does not prove, that the same codon-space descriptors operate at different biological layers in evolutionary versus engineering contexts. Family-boundary crossing is a real structural distinction over evolutionary timescales but did not predict acute transcriptomic perturbation in the Syn57 contrast; local neighborhood mismatch aligned with established recoding-burden covariates in the topology-fixed Napolitano arginine setting; and realized design deviations from the Syn57 design were dominated by mutational accessibility rather than systematic optimization of mismatch cost. We frame these as a working three-layer hypothesis (§3.9.2) suitable for prospective follow-up rather than as a confirmed model. Natural code evolution itself appears constrained along two partly independent axes—physicochemical smoothness and codon-family topological integrity—both of which sit above the layer at which acute synthetic-recoding burden manifests.

The framework's algebraic ambitions remain bounded. The KRAS–Fano clinical prediction, Serine distance-4 invariant, PSL(2,7) symmetry, and holomorphic-embedding claims were tested and rejected (Table 1) and are out of scope. The $\mathrm{GF}(2)^6$ representation contributes two genuine pieces of analytical machinery—the ρ-sweep that interpolates between Q_6 and $H(3,4)$, and the ε-filtration that gives codon-family connectivity a persistence parameter—plus a third, the encoding-sensitivity audit, that is best understood as methodological hygiene the binary representation owes the framework rather than as a contribution from it. What survives is a measured, encoding-independent picture of the genetic code as a low-cost coloring of $H(3,4)$ whose persistent reassignment trajectories respect both the physicochemistry of the substituted amino acid and the connectivity of the codon family it joins.

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Data availability

All code, raw data, and intermediate analysis outputs are publicly released in the `codontopo` repository at https://github.com/biostochastics/codontopo (version 0.4.0, commit `0ab163d`). Install with `pip install -e ".[all]"` or `uv sync --all-extras`. Analyses are fully reproducible via:

```
git clone https://github.com/biostochastics/codontopo.git
cd codontopo
codon-topo all --output-dir=./output --seed=135325
```

All inline statistics, tables, and figures in this manuscript and supplement are rendered directly from the pipeline outputs by the Typst sources (`manuscript.typ`, `supplement.typ`) included in the same repository. NCBI genome assembly accessions are listed in Table 8.

Declaration of competing interest

The authors declare no competing interests.

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